

BFS Traversal

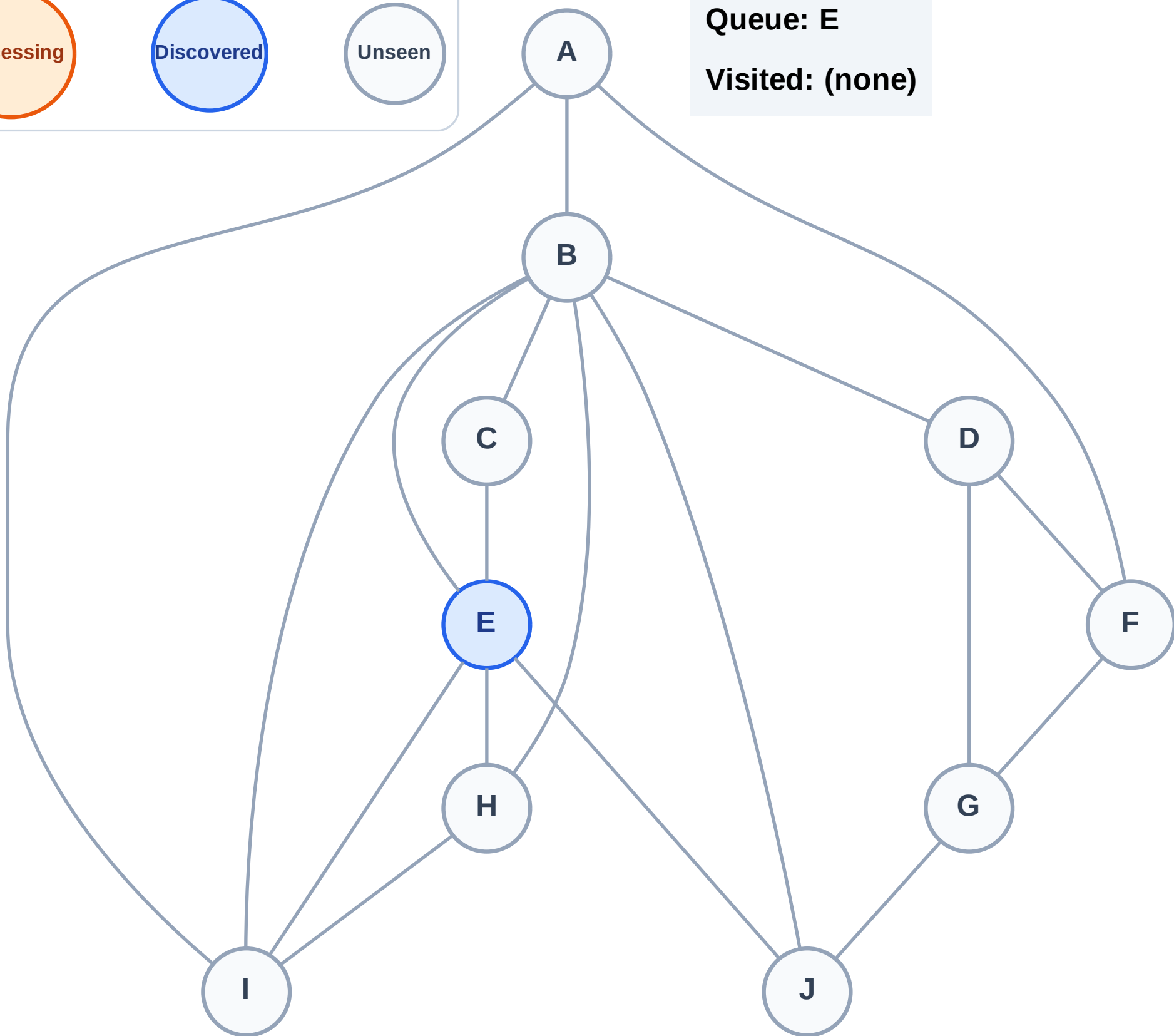
Step 1: Start at E

Legend



Queue: E

Visited: (none)



BFS Traversal

Step 2: Process node E

Legend

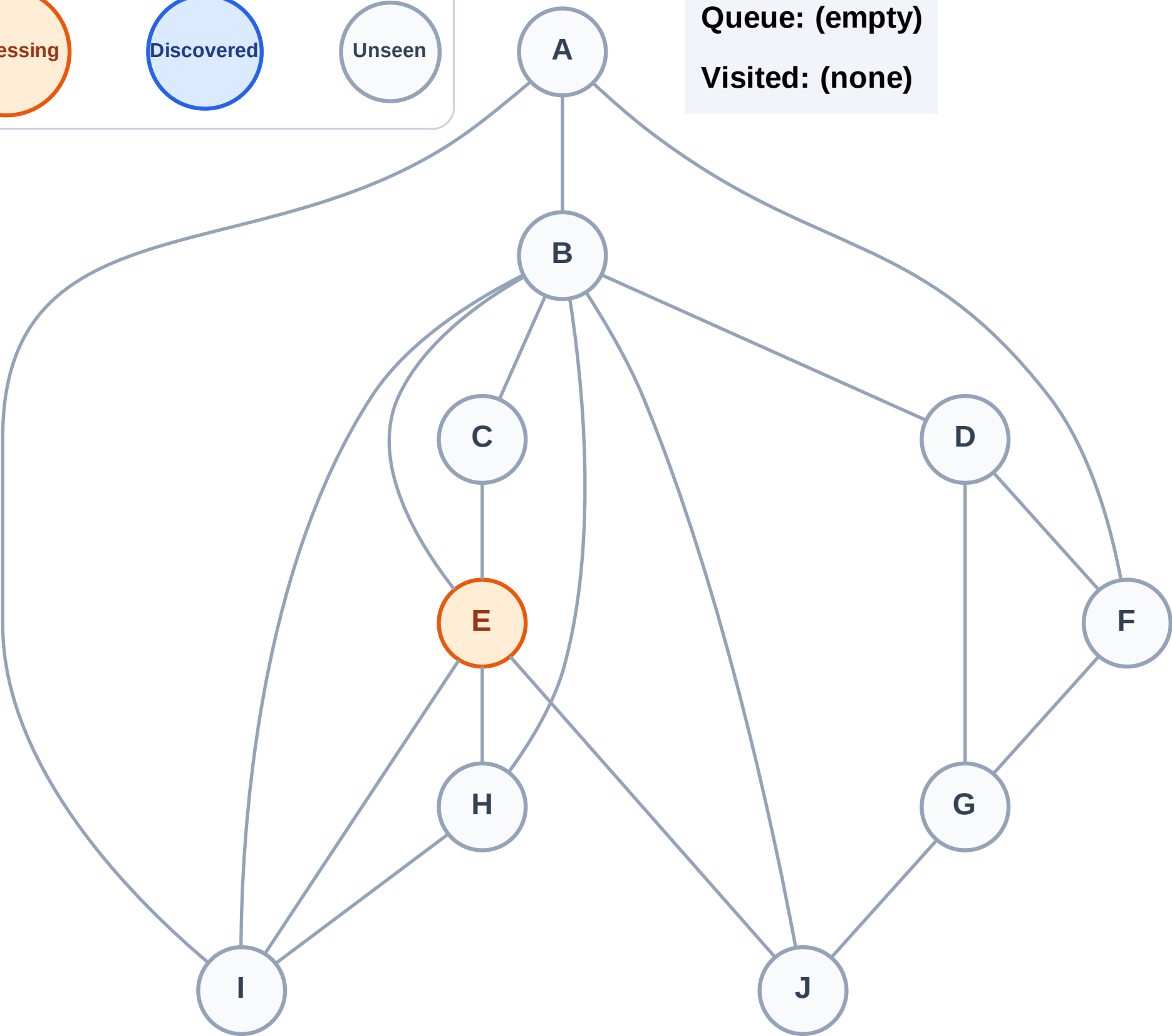
Complete

Processing

Discovered

Unseen

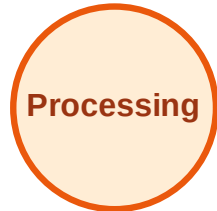
Queue: (empty)
Visited: (none)



BFS Traversal

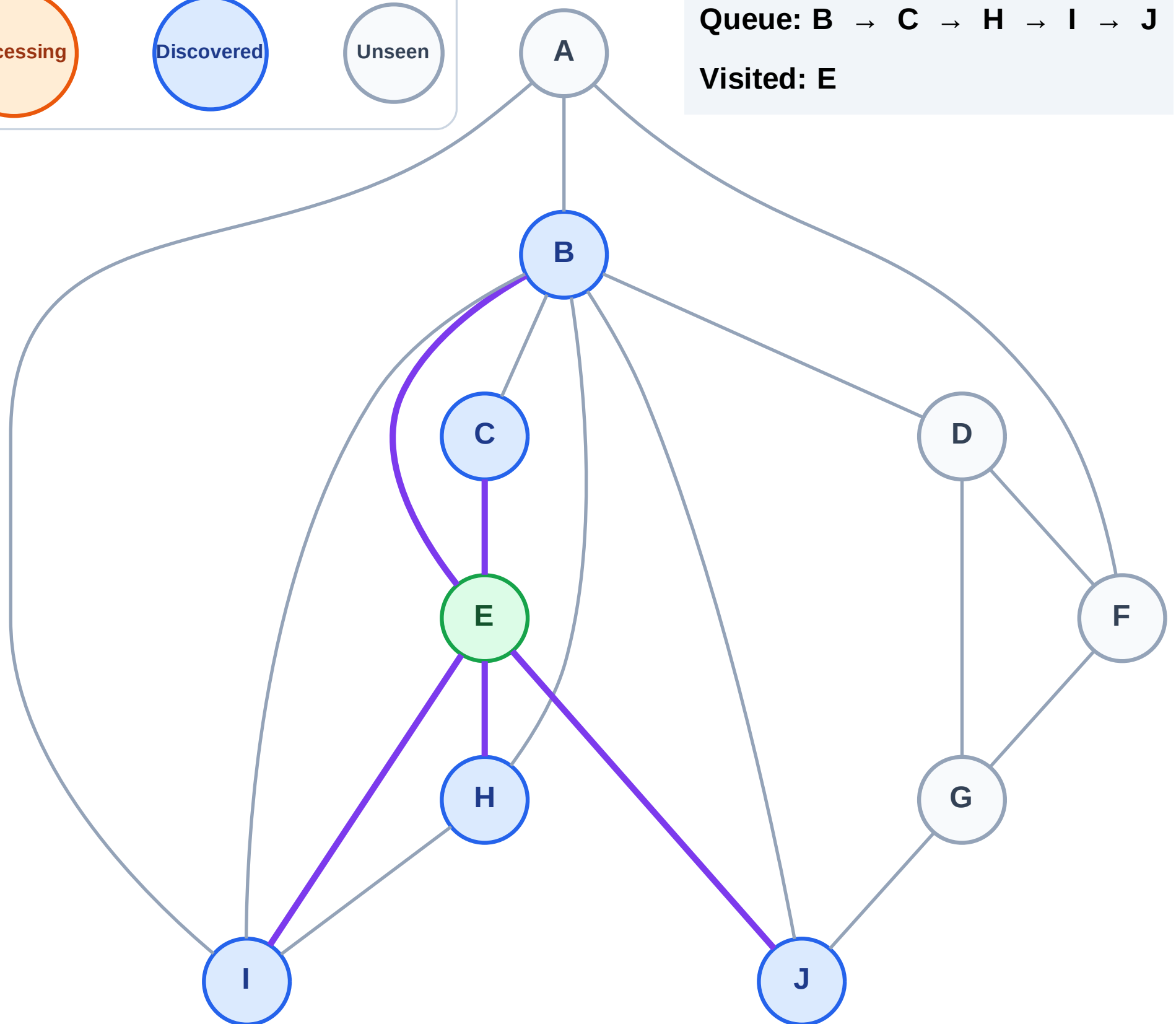
Step 3: Discover B, C, H, I, J from E

Legend



Queue: B → C → H → I → J

Visited: E



BFS Traversal

Step 4: Process node B

Legend

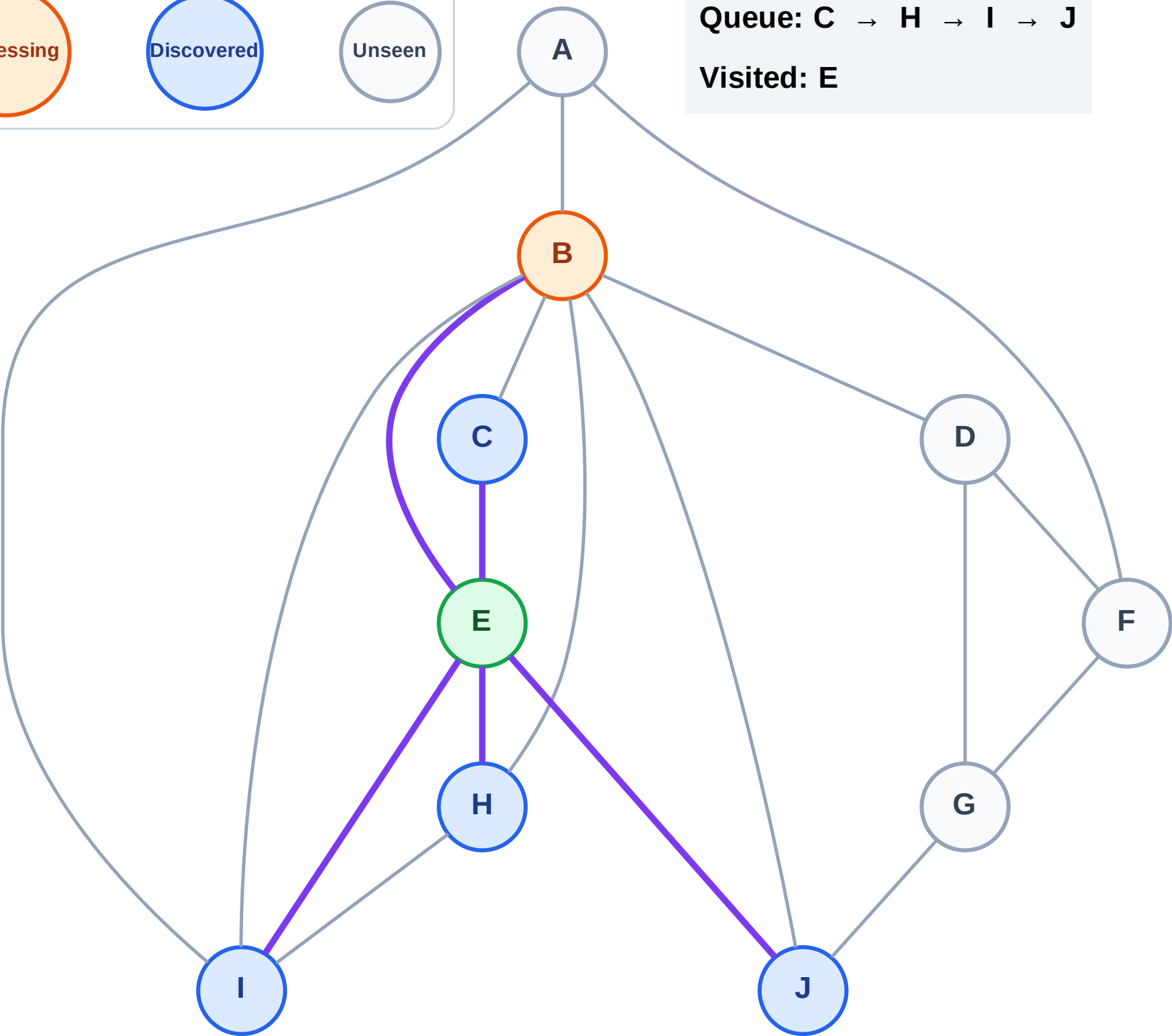
Complete

Processing

Discovered

Unseen

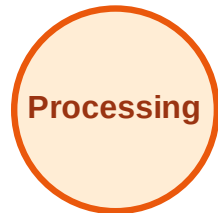
Queue: C → H → I → J
Visited: E



BFS Traversal

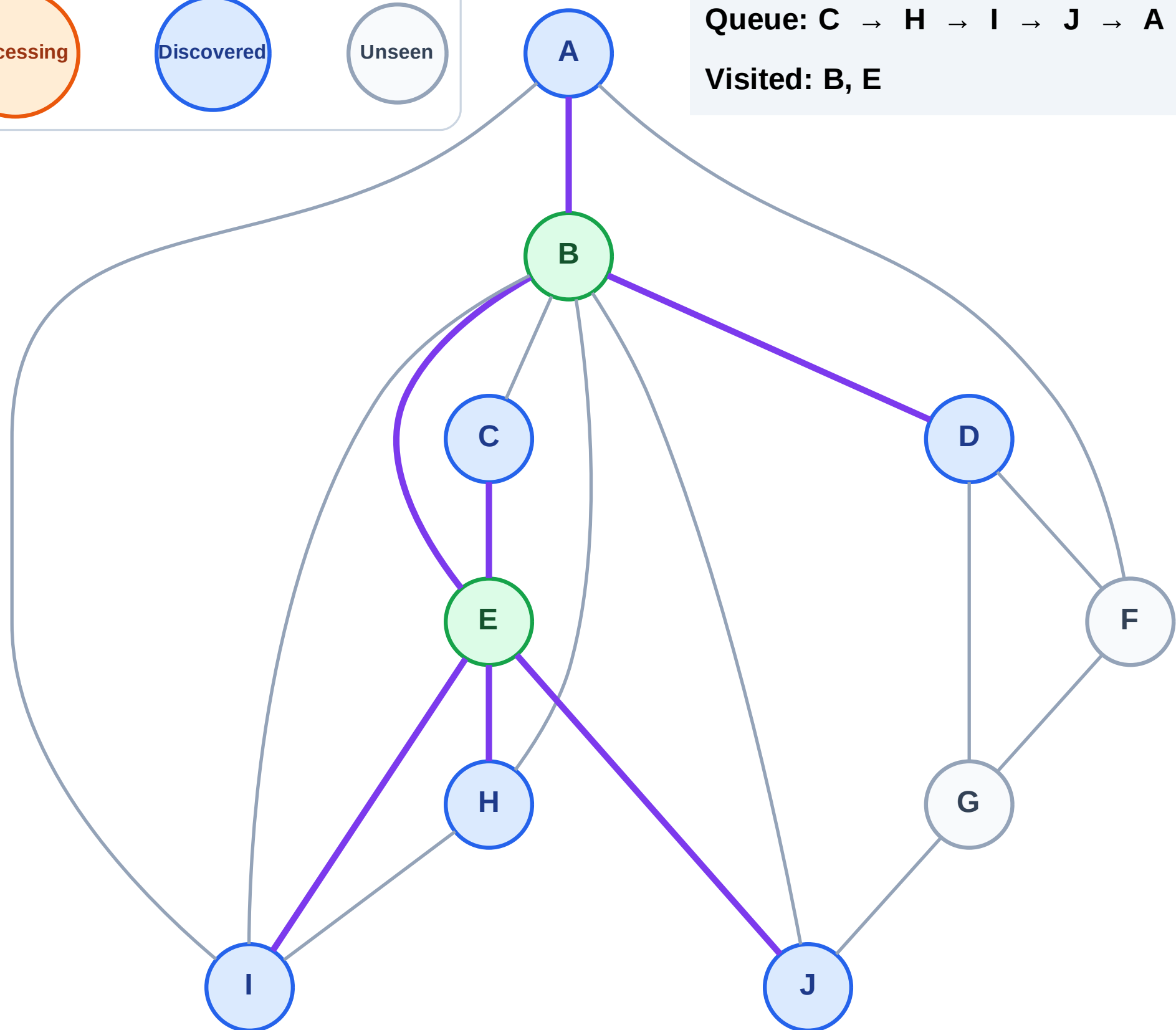
Step 5: Discover A, D from B

Legend



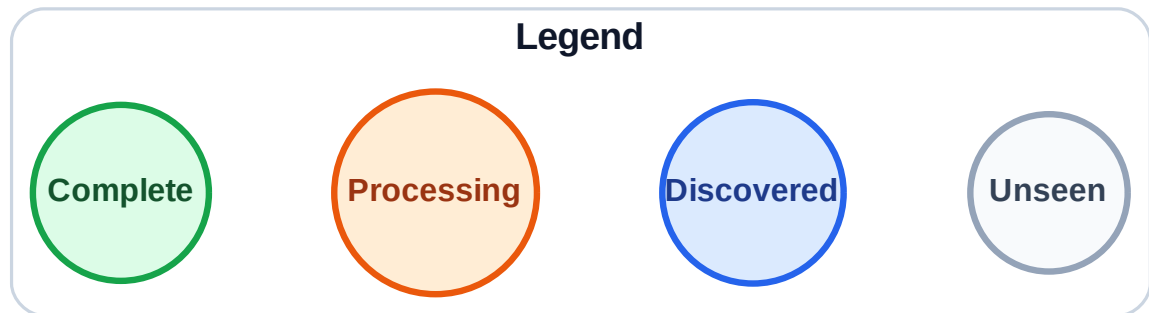
Queue: C → H → I → J → A → D

Visited: B, E



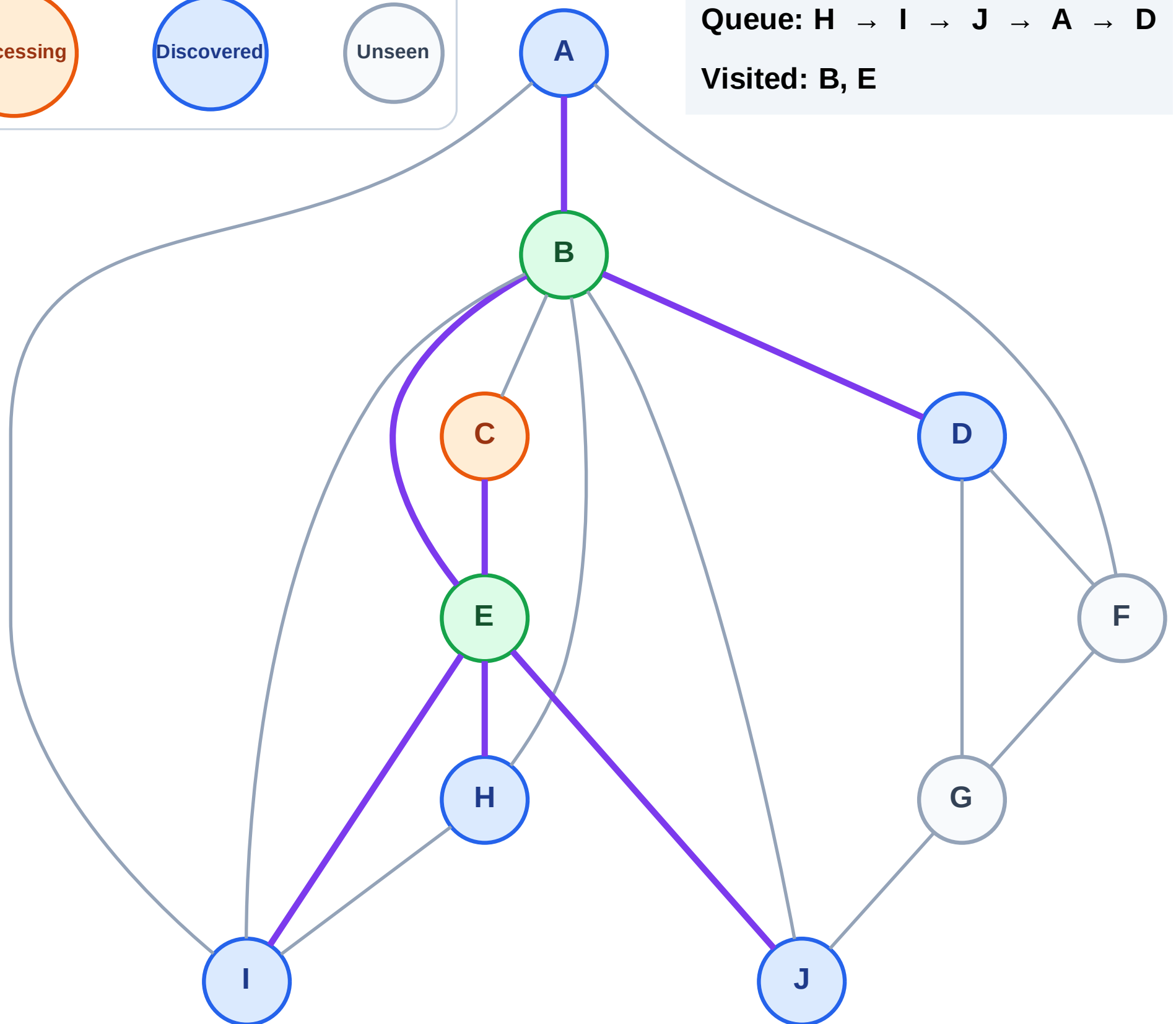
Step 6: Process node C

Step 6: Process node C



Queue: H → I → J → A → D

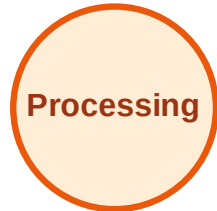
Visited: B, E



BFS Traversal

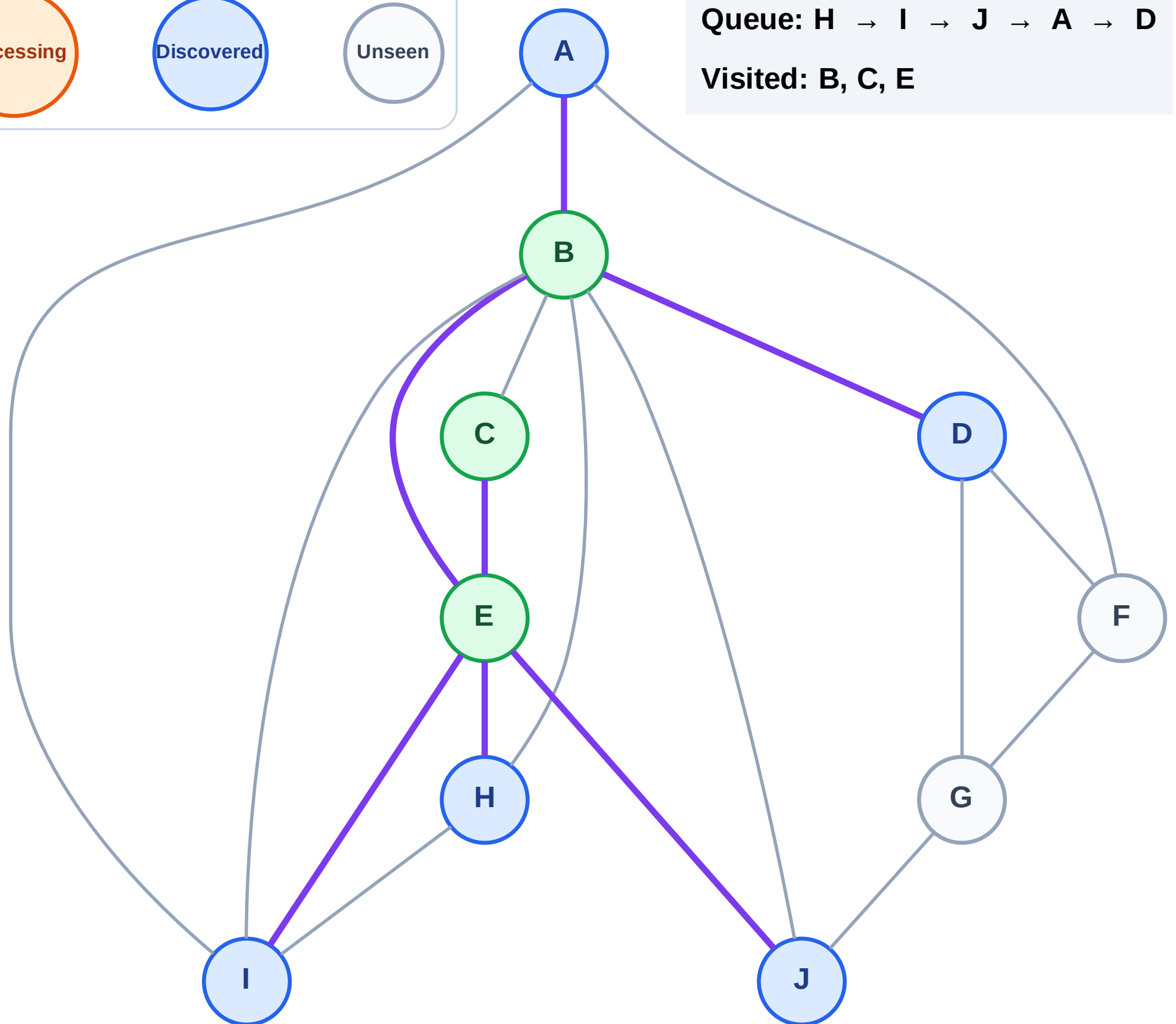
Step 7: No new neighbors from C

Legend



Queue: H → I → J → A → D

Visited: B, C, E



BFS Traversal

Step 8: Process node H

Legend

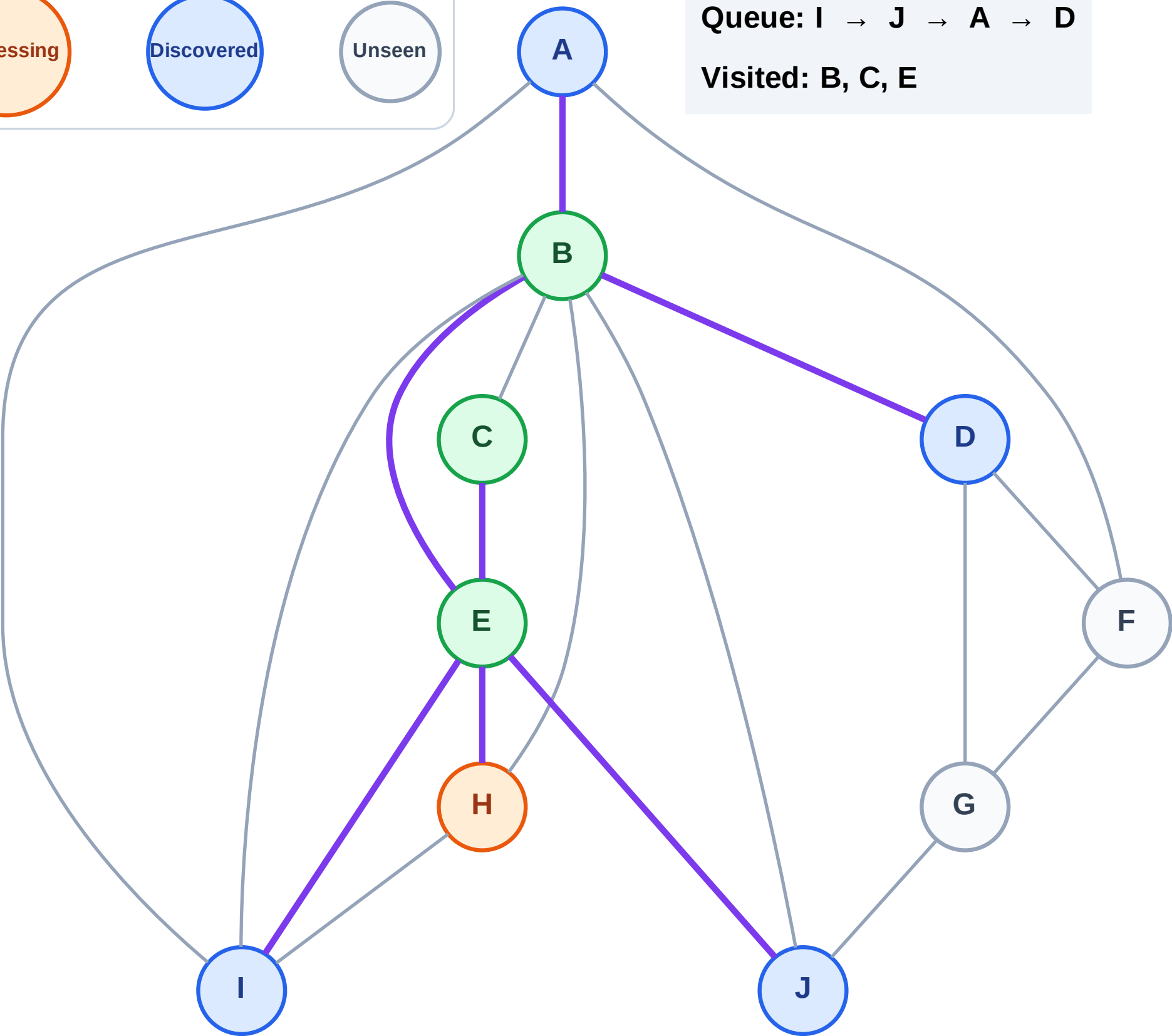
Complete

Processing

Discovered

Unseen

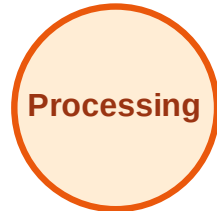
Queue: I → J → A → D
Visited: B, C, E



BFS Traversal

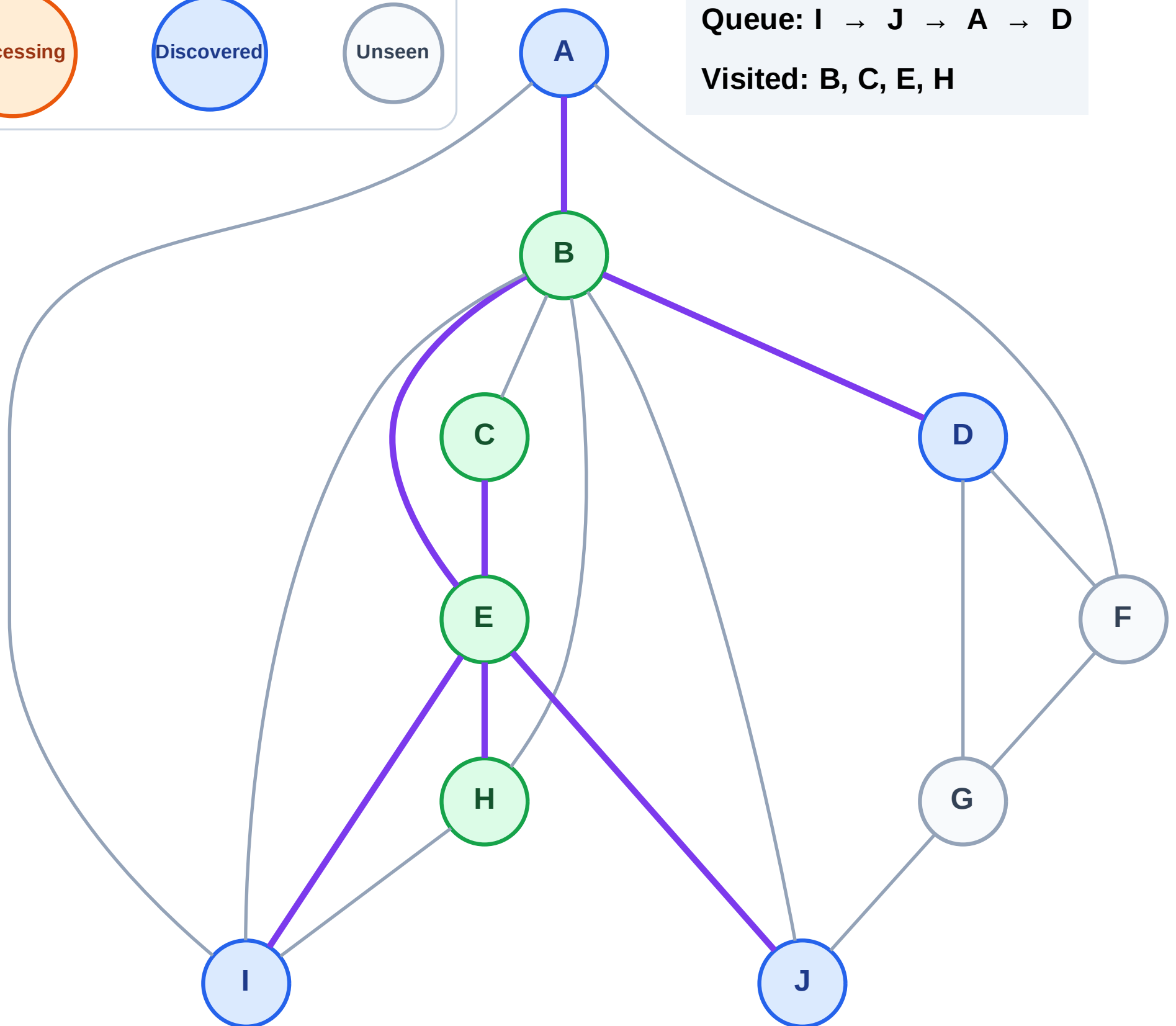
Step 9: No new neighbors from H

Legend



Queue: I → J → A → D

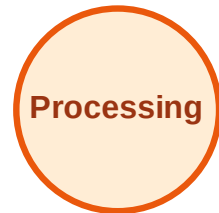
Visited: B, C, E, H



BFS Traversal

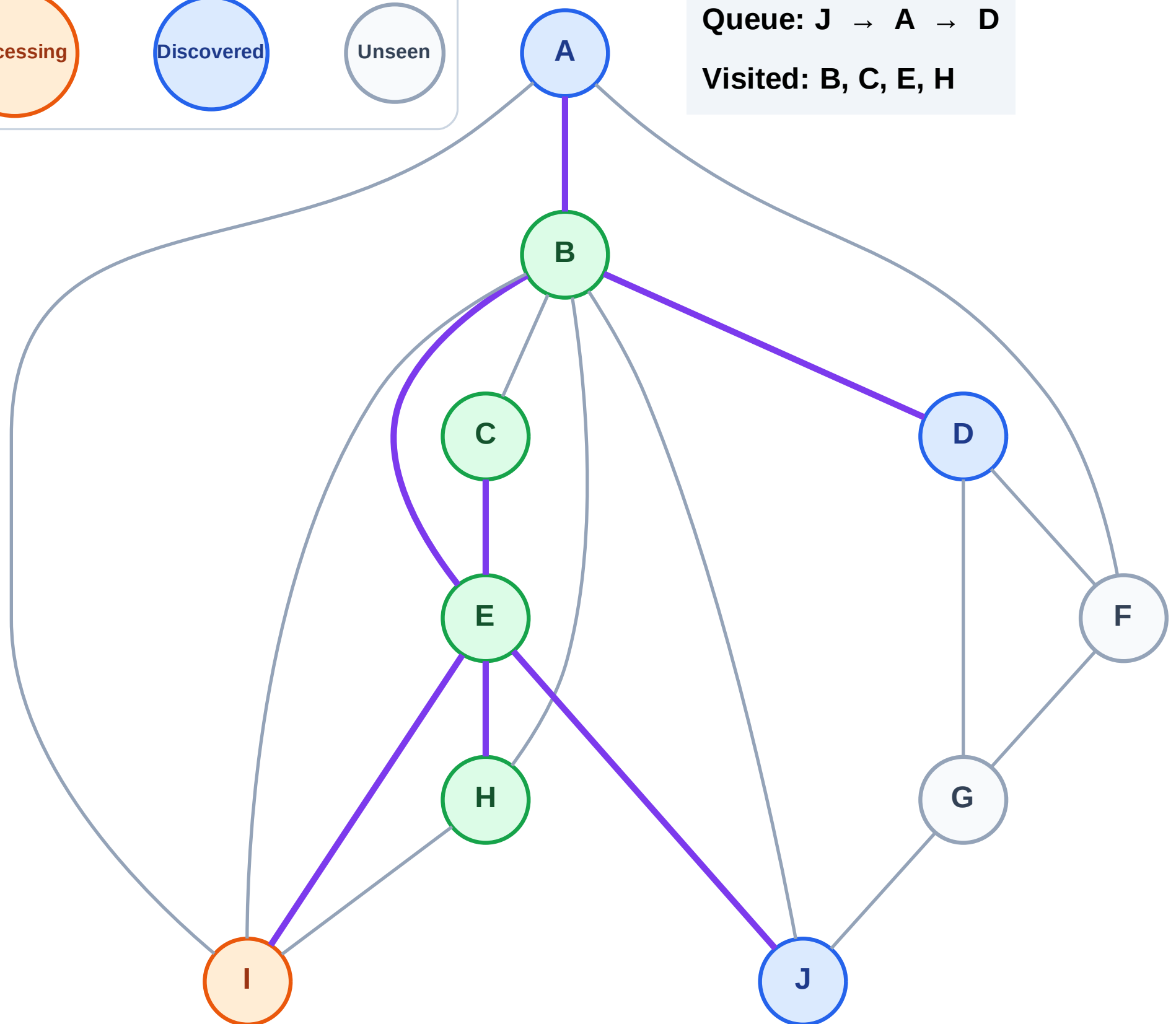
Step 10: Process node I

Legend



Queue: J → A → D

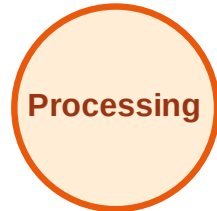
Visited: B, C, E, H



BFS Traversal

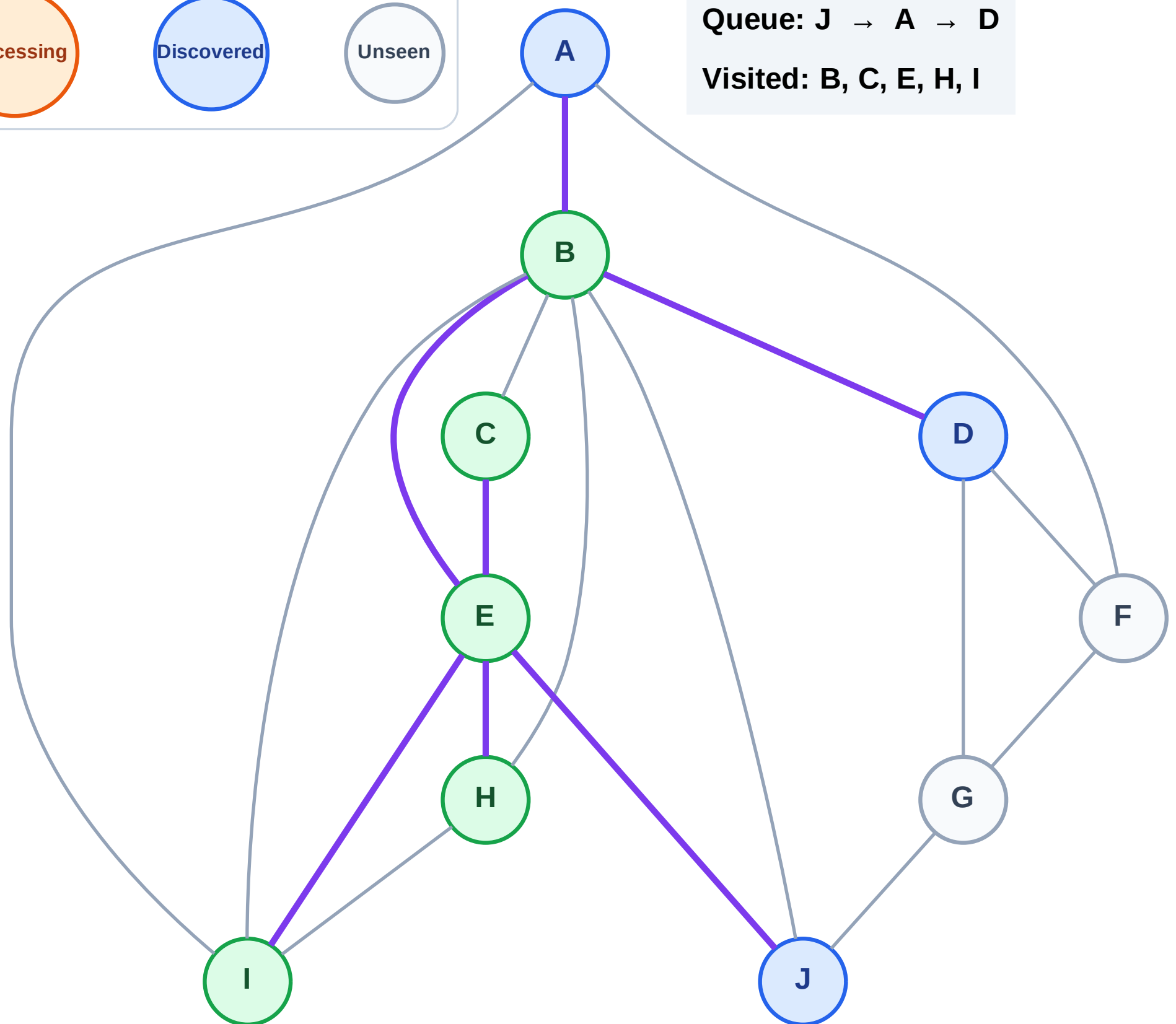
Step 11: No new neighbors from I

Legend



Queue: J → A → D

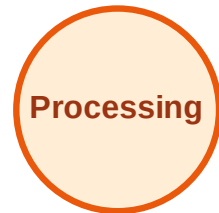
Visited: B, C, E, H, I



BFS Traversal

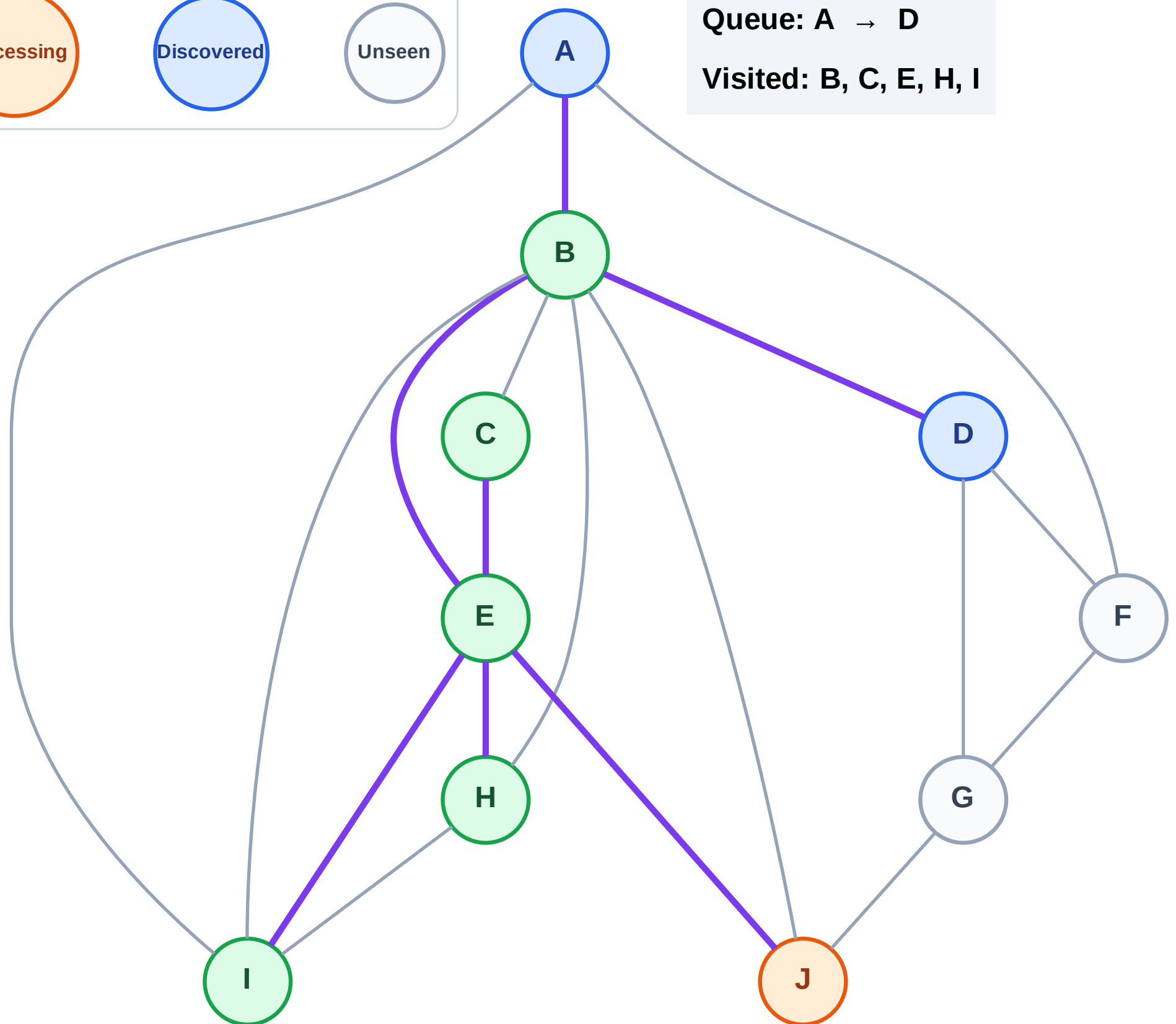
Step 12: Process node J

Legend



Queue: A → D

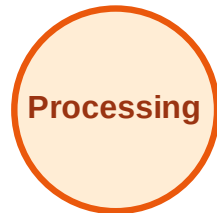
Visited: B, C, E, H, I



BFS Traversal

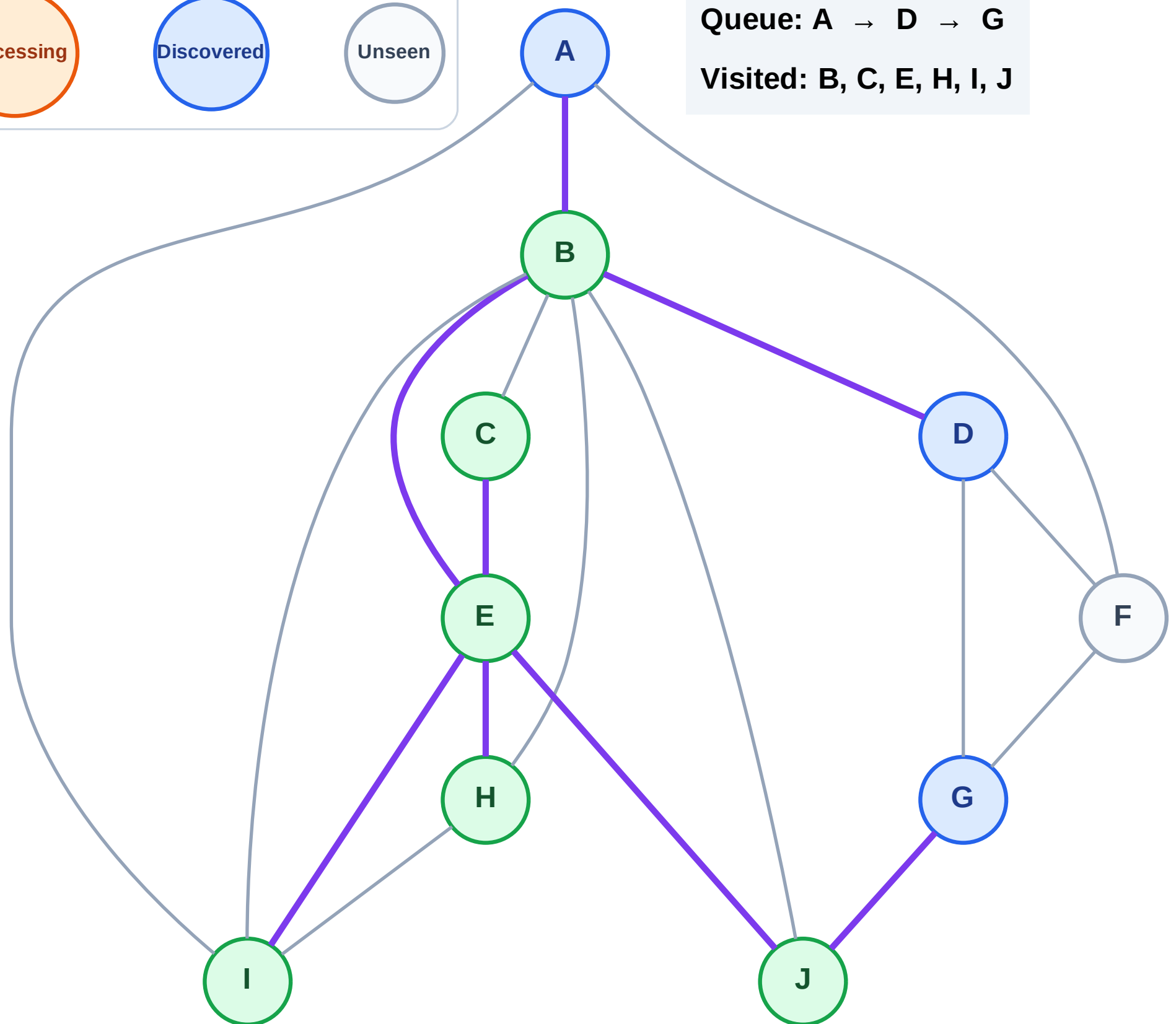
Step 13: Discover G from J

Legend



Queue: A → D → G

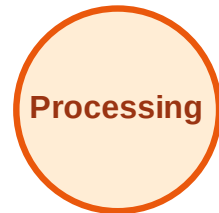
Visited: B, C, E, H, I, J



BFS Traversal

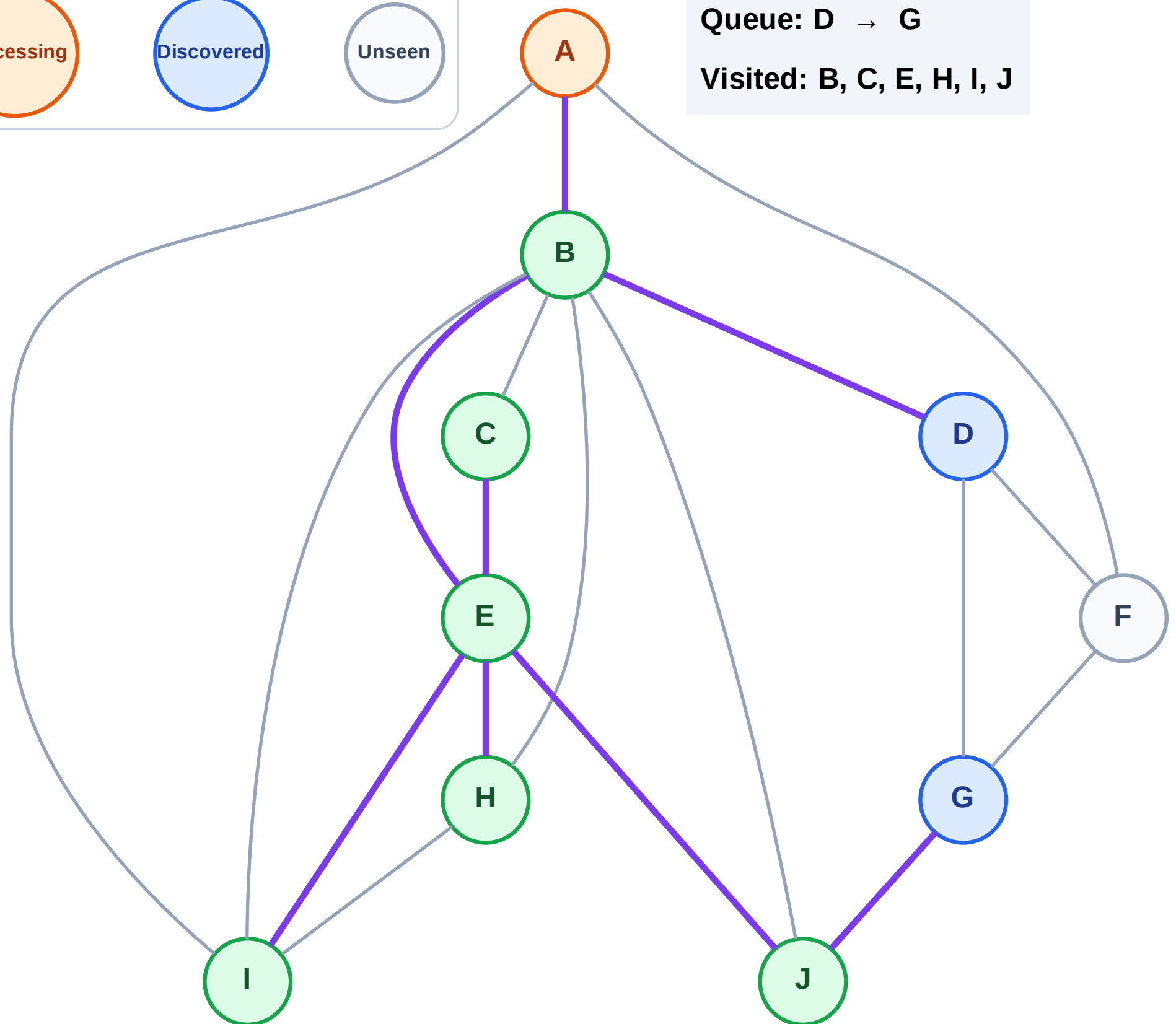
Step 14: Process node A

Legend



Queue: D → G

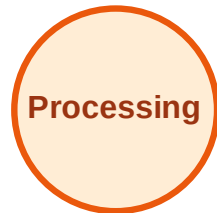
Visited: B, C, E, H, I, J



BFS Traversal

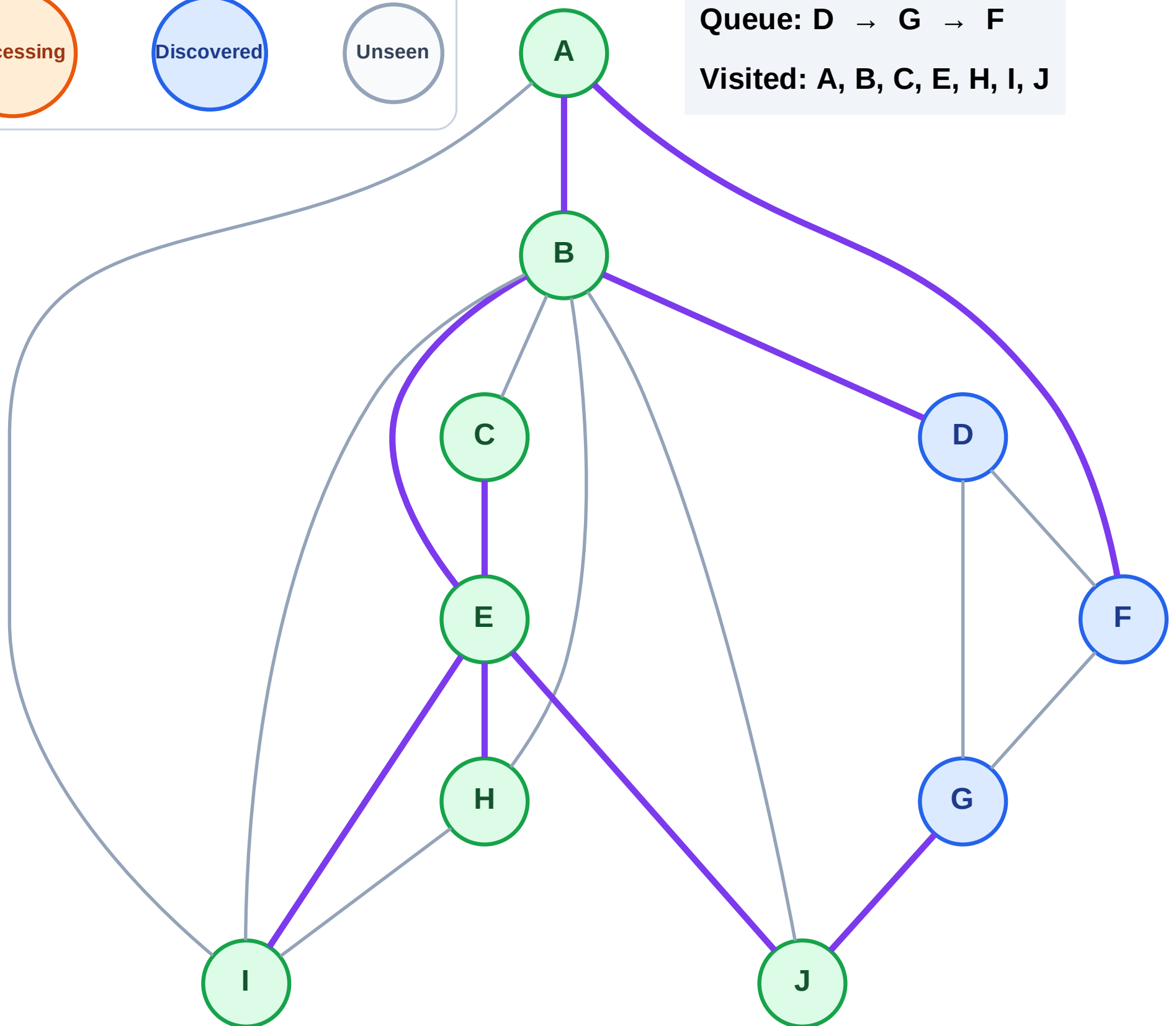
Step 15: Discover F from A

Legend



Queue: D → G → F

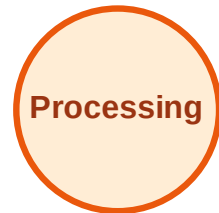
Visited: A, B, C, E, H, I, J



BFS Traversal

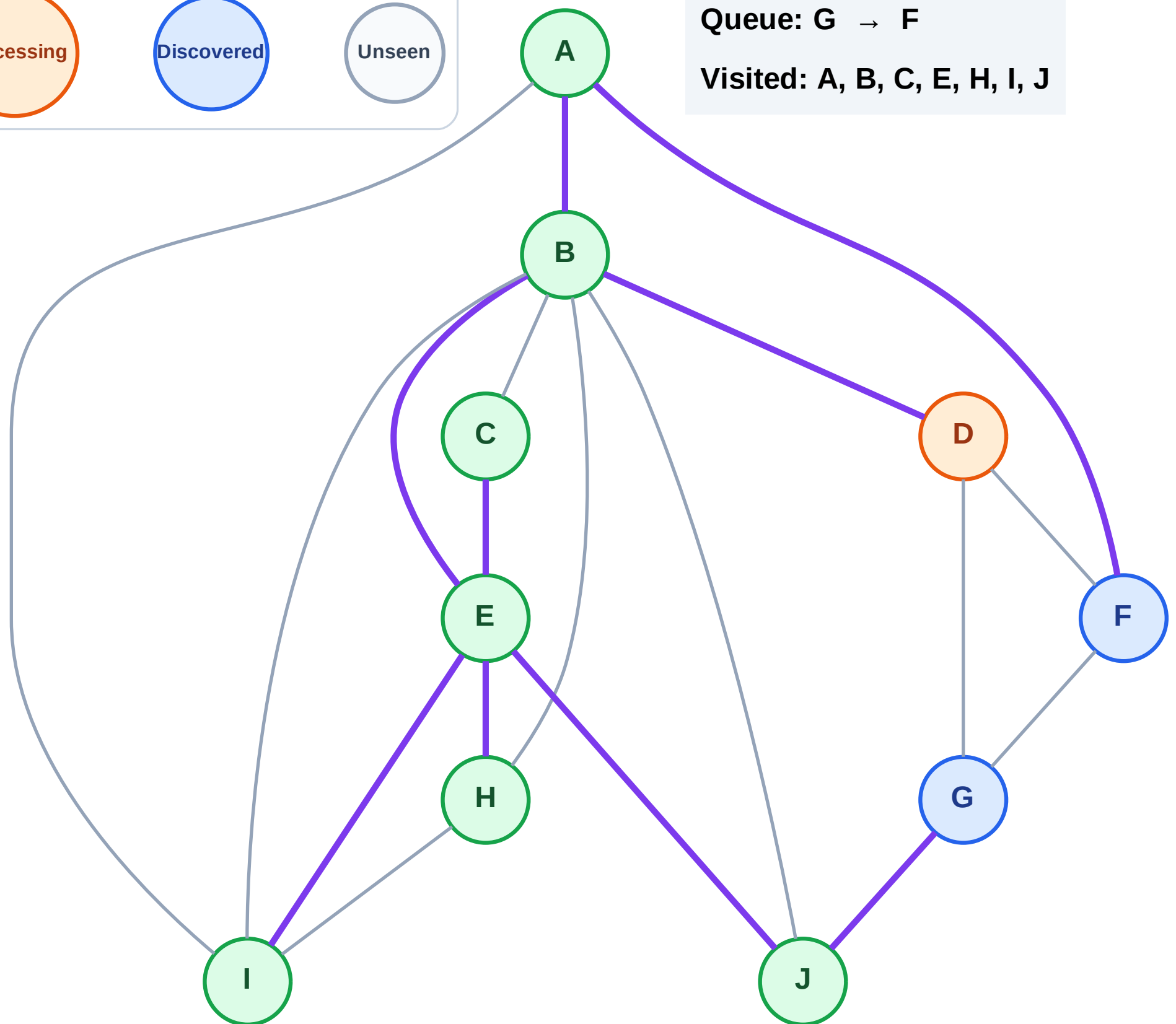
Step 16: Process node D

Legend



Queue: G → F

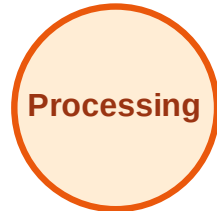
Visited: A, B, C, E, H, I, J



BFS Traversal

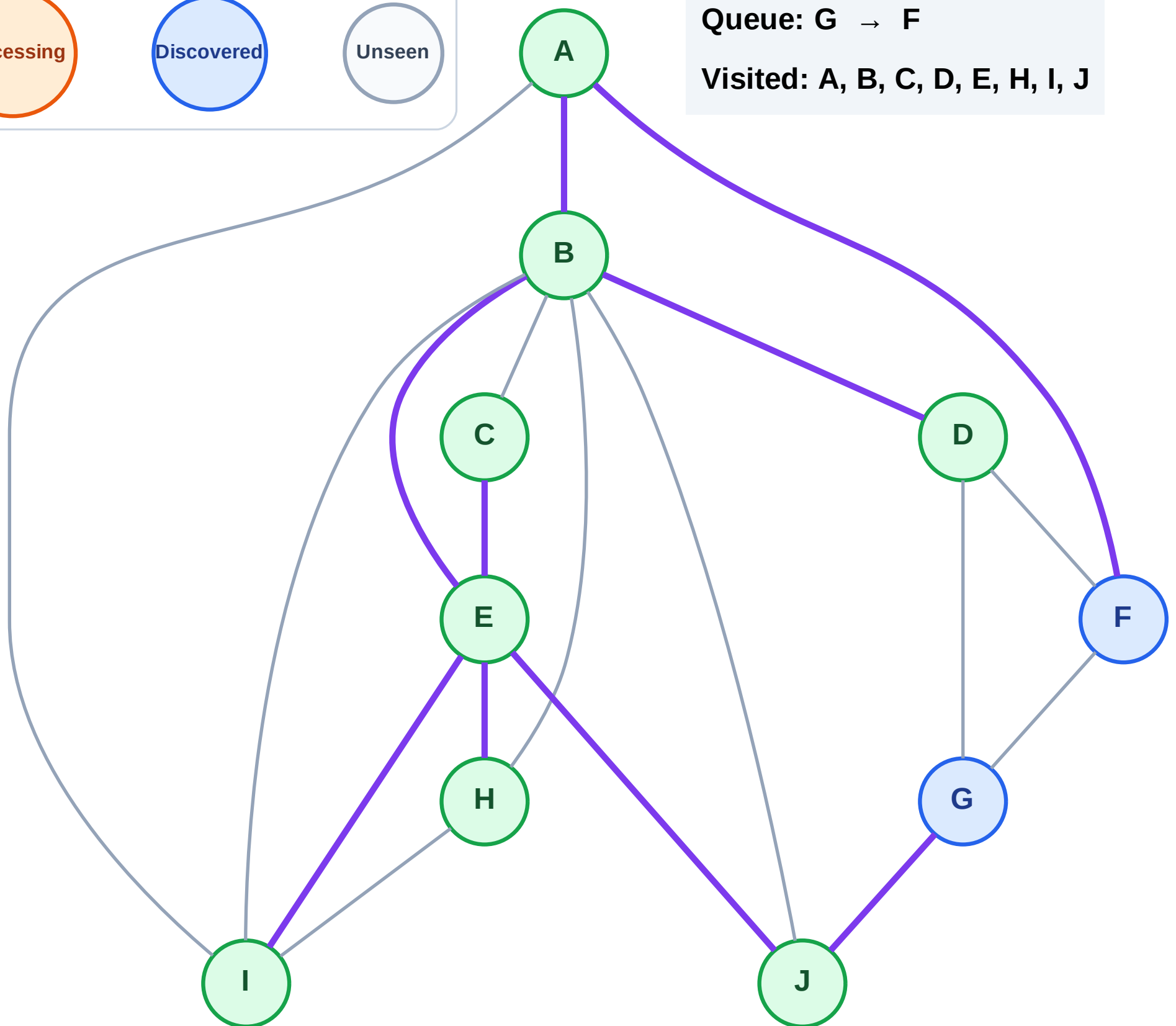
Step 17: No new neighbors from D

Legend



Queue: G → F

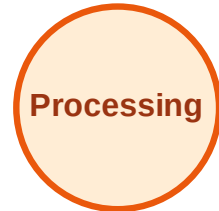
Visited: A, B, C, D, E, H, I, J



BFS Traversal

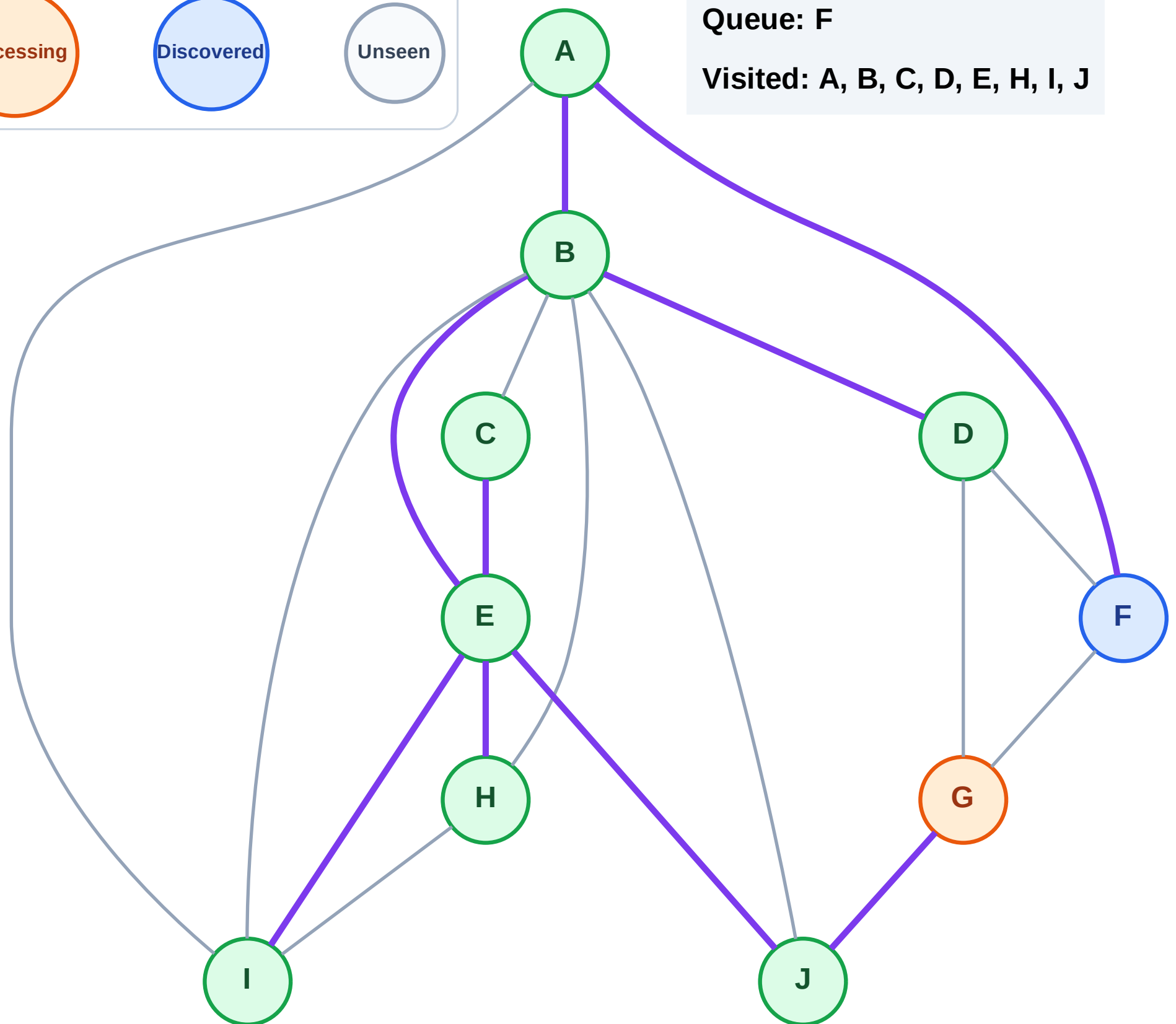
Step 18: Process node G

Legend



Queue: F

Visited: A, B, C, D, E, H, I, J



BFS Traversal

Step 19: No new neighbors from G

Legend

Complete

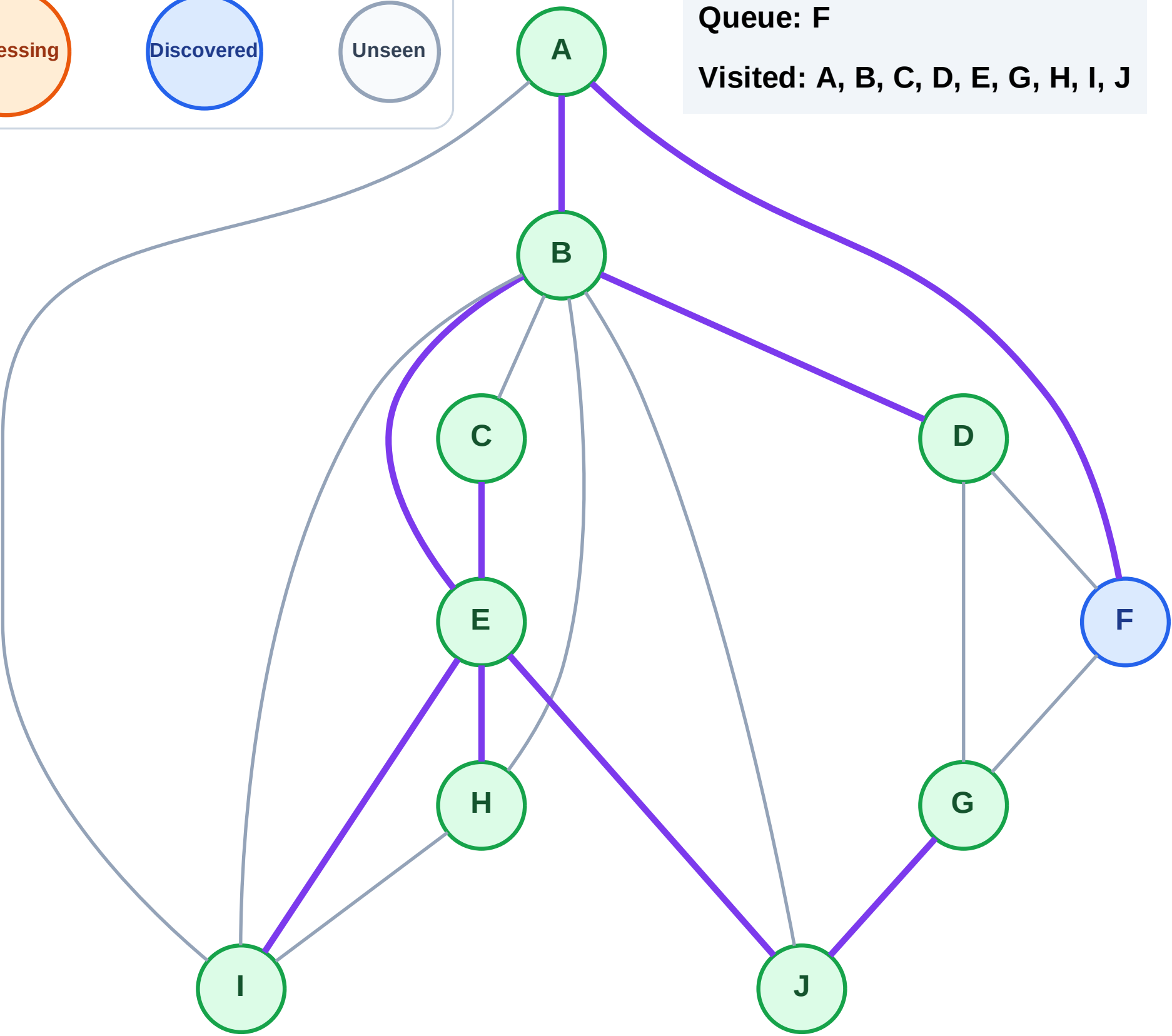
Processing

Discovered

Unseen

Queue: F

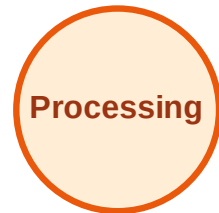
Visited: A, B, C, D, E, G, H, I, J



BFS Traversal

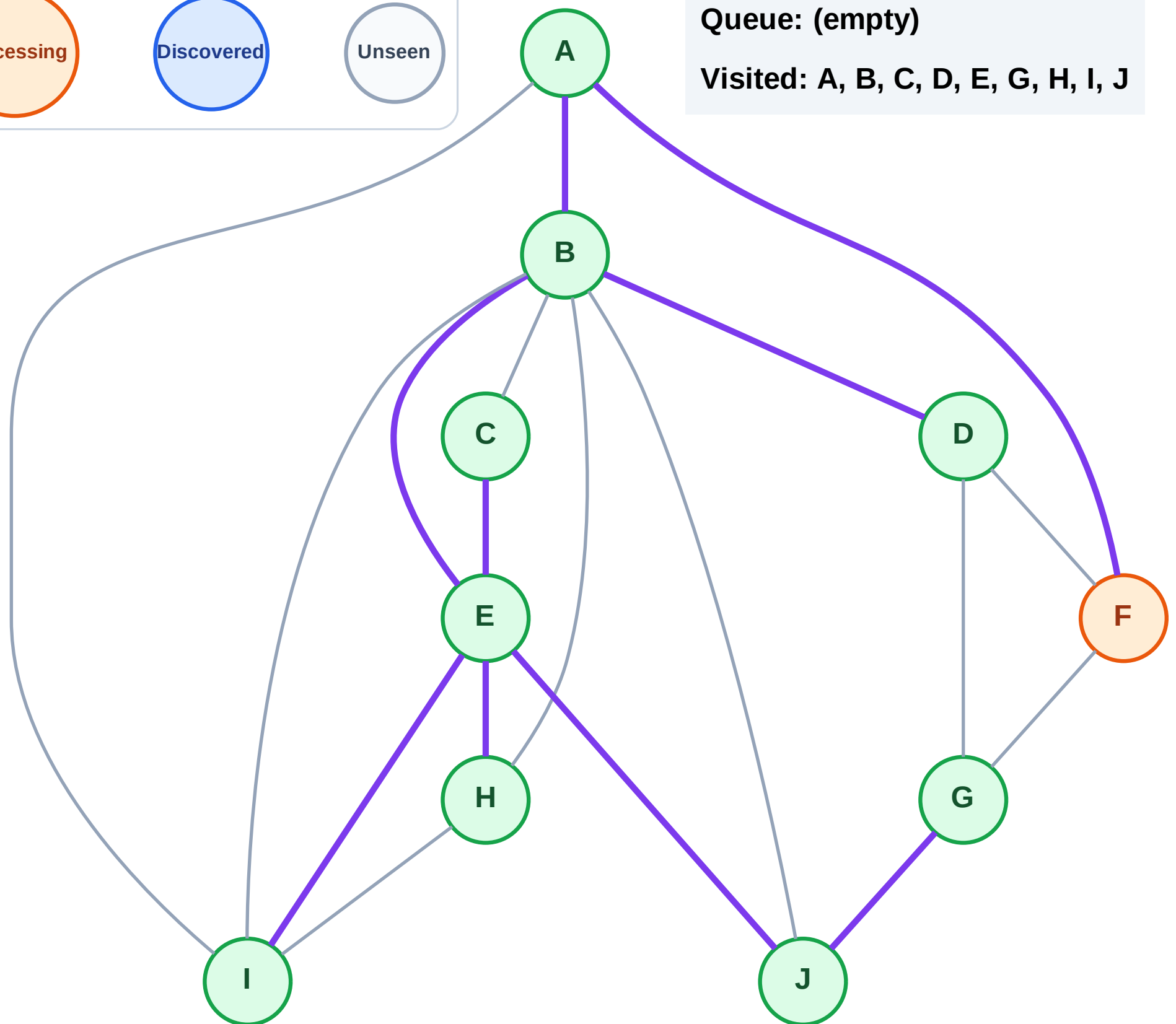
Step 20: Process node F

Legend



Queue: (empty)

Visited: A, B, C, D, E, G, H, I, J



BFS Traversal

Step 21: No new neighbors from F

Legend

Complete

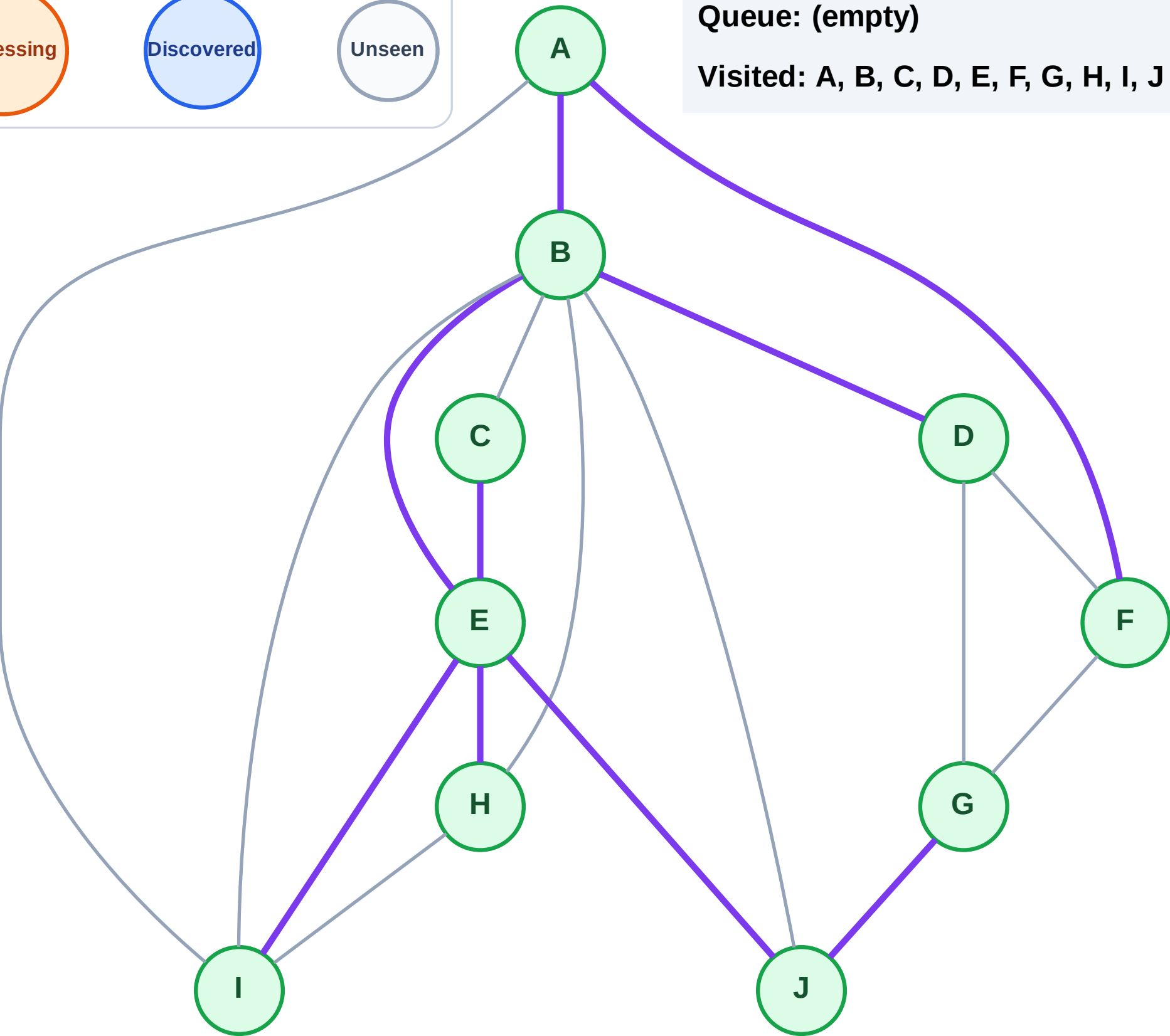
Processing

Discovered

Unseen

Queue: (empty)

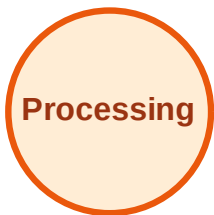
Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

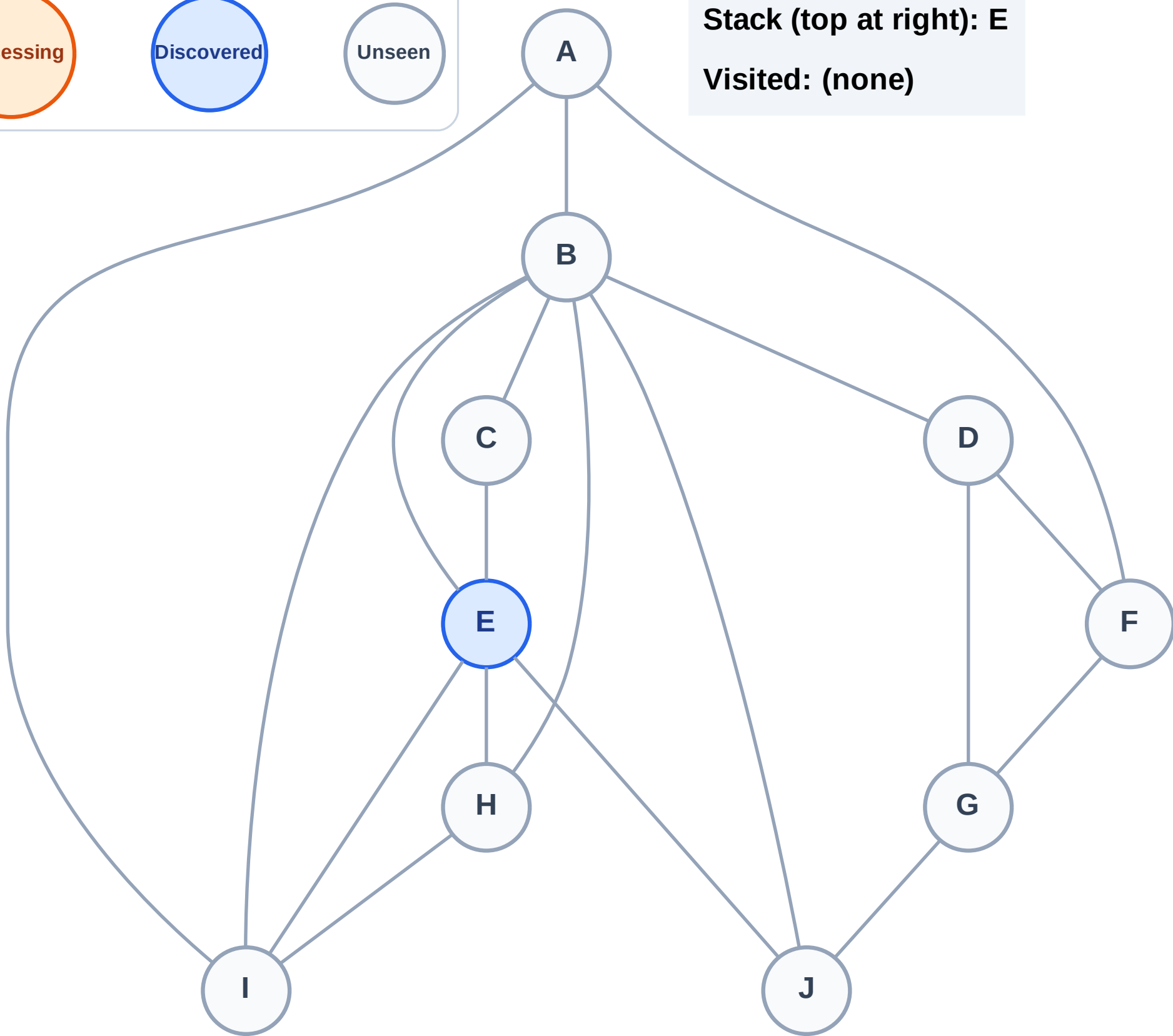
Step 1: Start at E

Legend



Stack (top at right): E

Visited: (none)



DFS Traversal

Step 2: Process node E

Legend

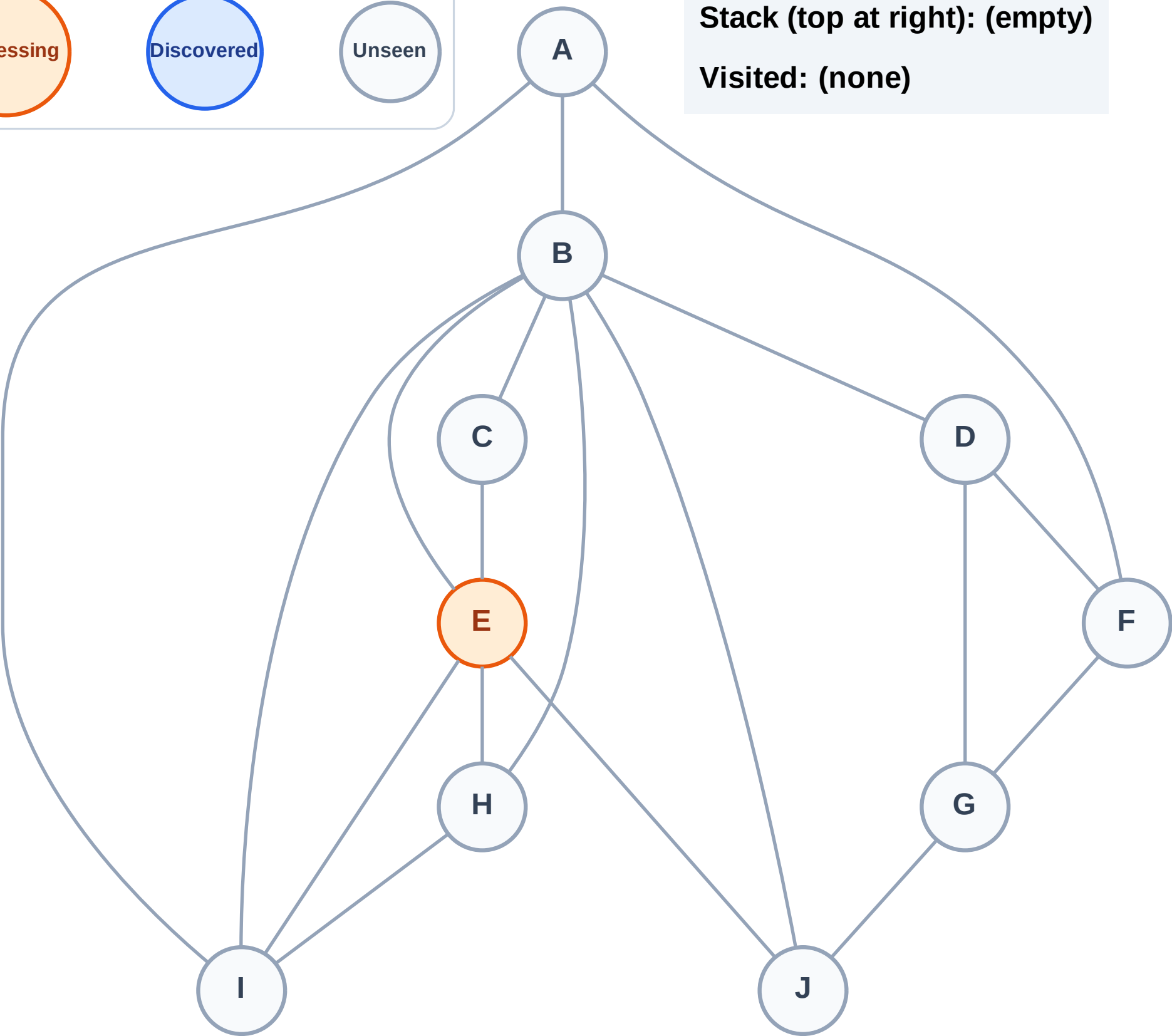
Complete

Processing

Discovered

Unseen

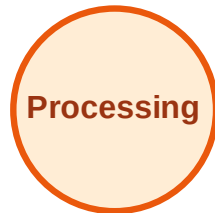
Stack (top at right): (empty)
Visited: (none)



DFS Traversal

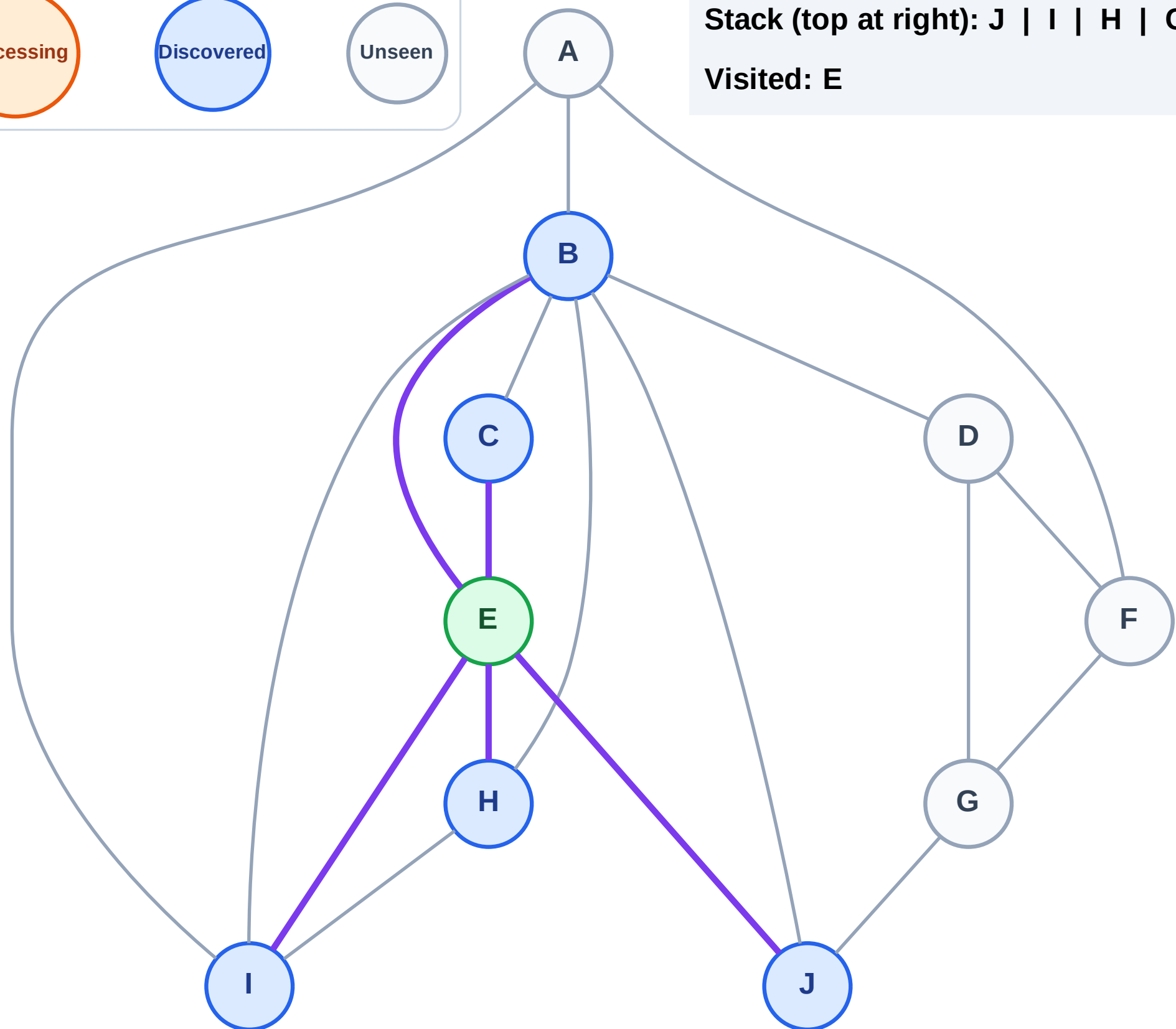
Step 3: Discover J, I, H, C, B from E

Legend



Stack (top at right): J | I | H | C | B

Visited: E



DFS Traversal

Step 4: Process node B

Legend

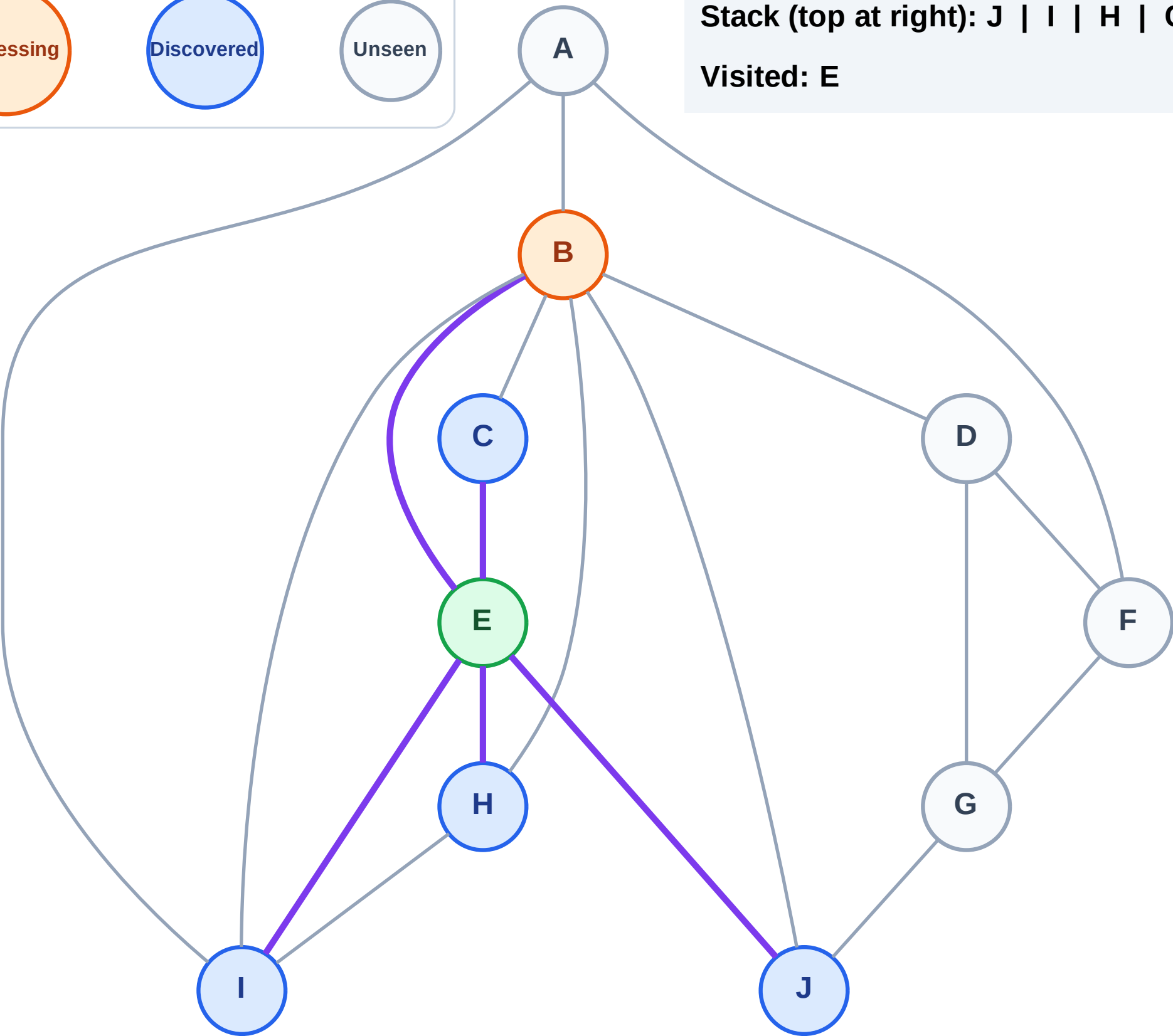
Complete

Processing

Discovered

Unseen

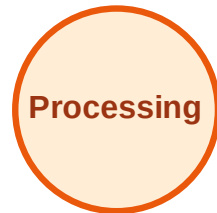
Stack (top at right): J | I | H | C
Visited: E



DFS Traversal

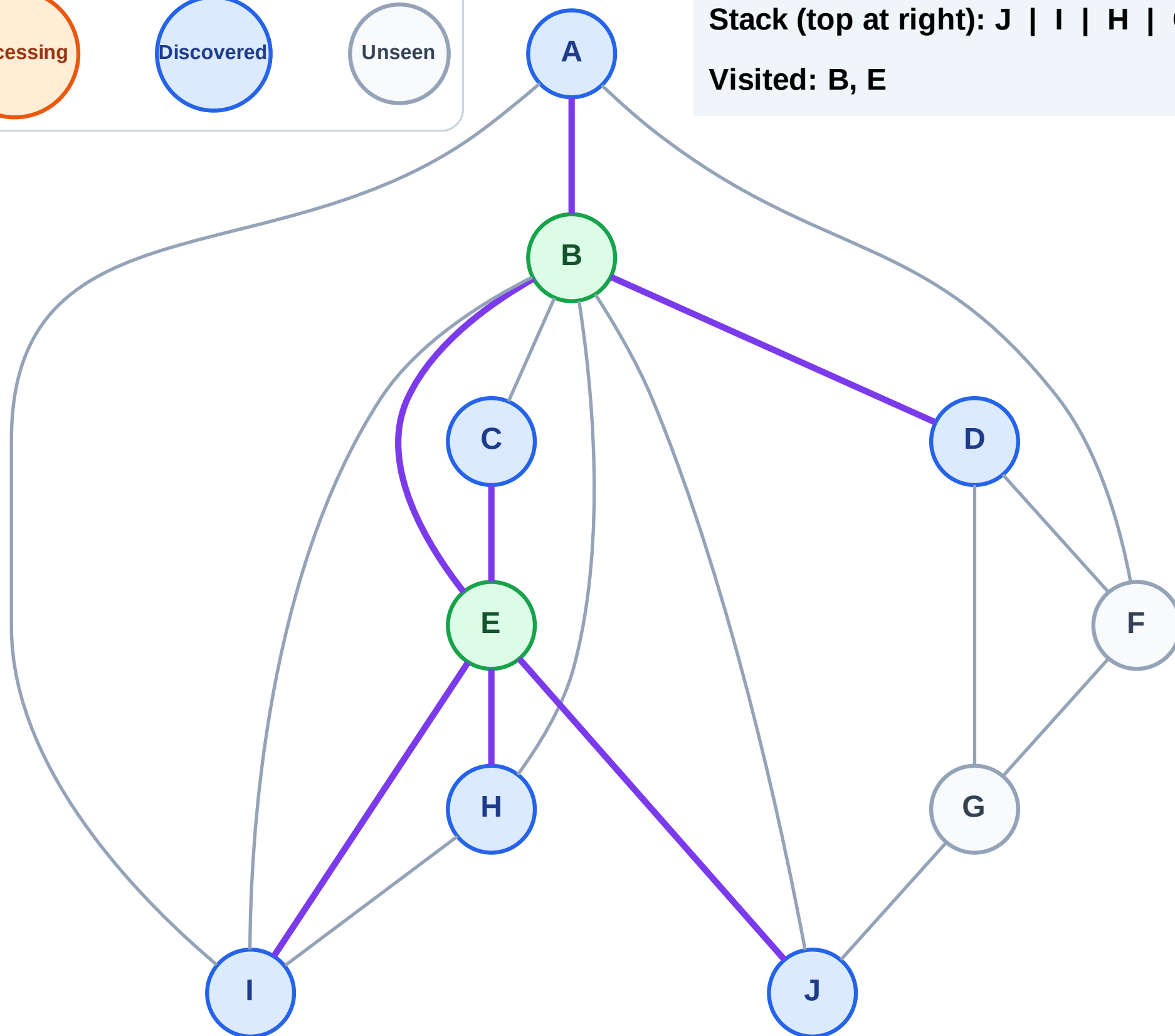
Step 5: Discover D, A from B

Legend



Stack (top at right): J | I | H | C | D | A

Visited: B, E



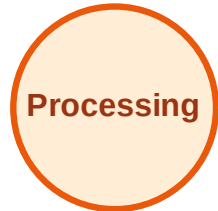
DFS Traversal

Step 6: Process node A

Legend



Complete



Processing



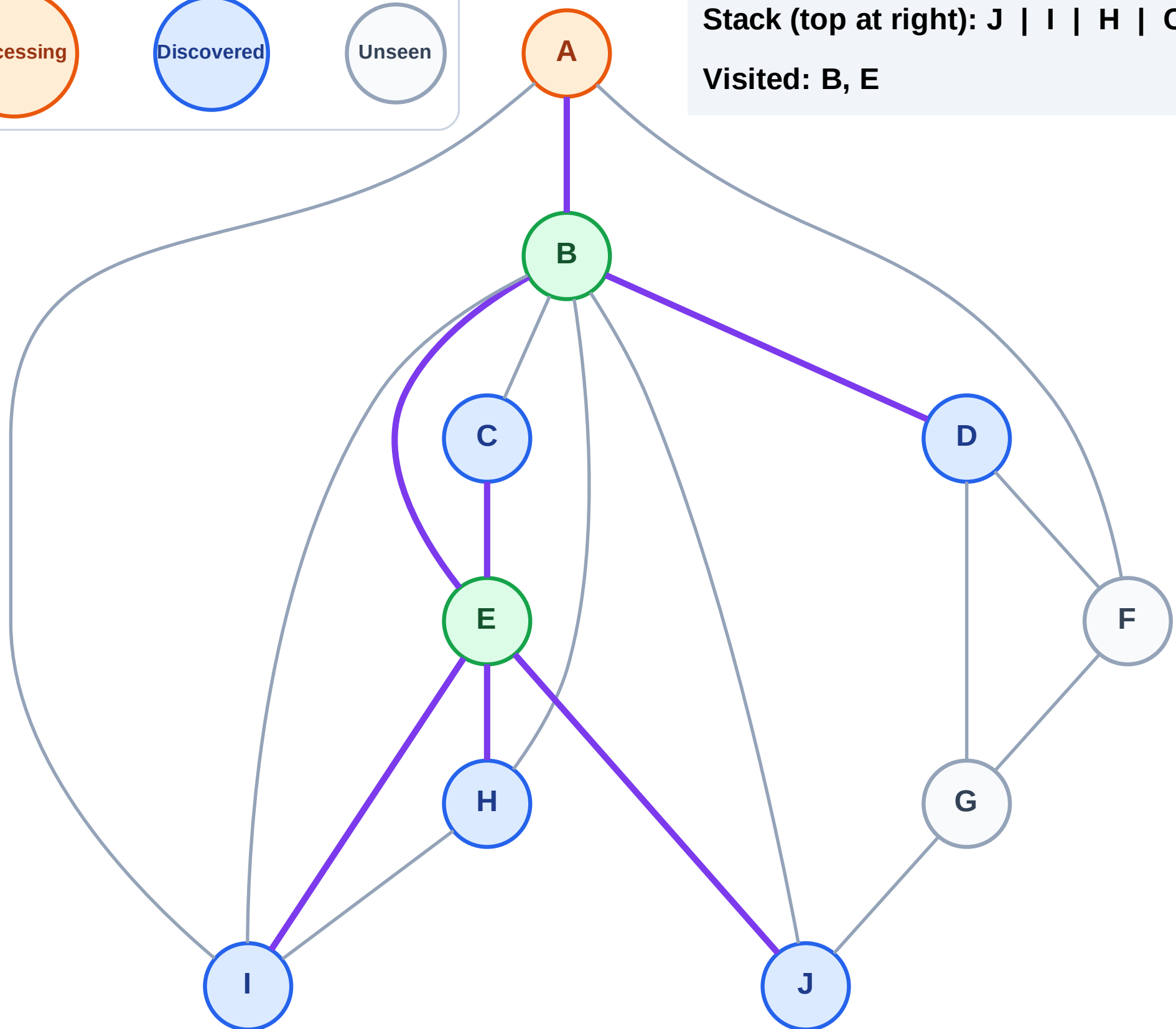
Discovered



Unseen

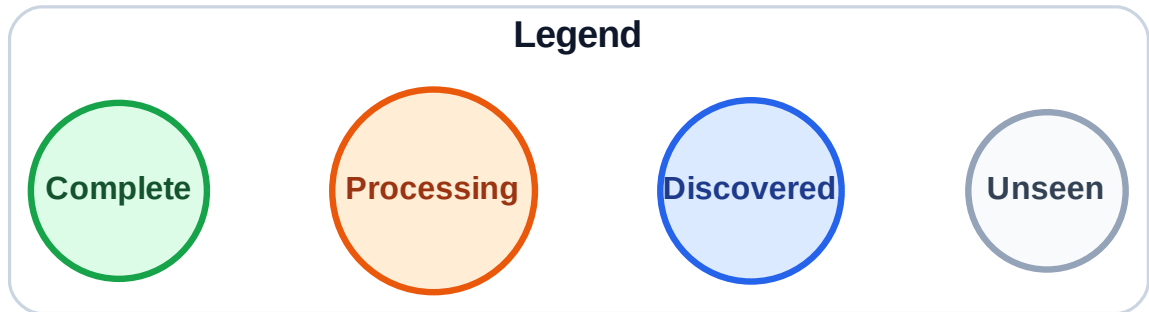
Stack (top at right): J | I | H | C | D

Visited: B, E

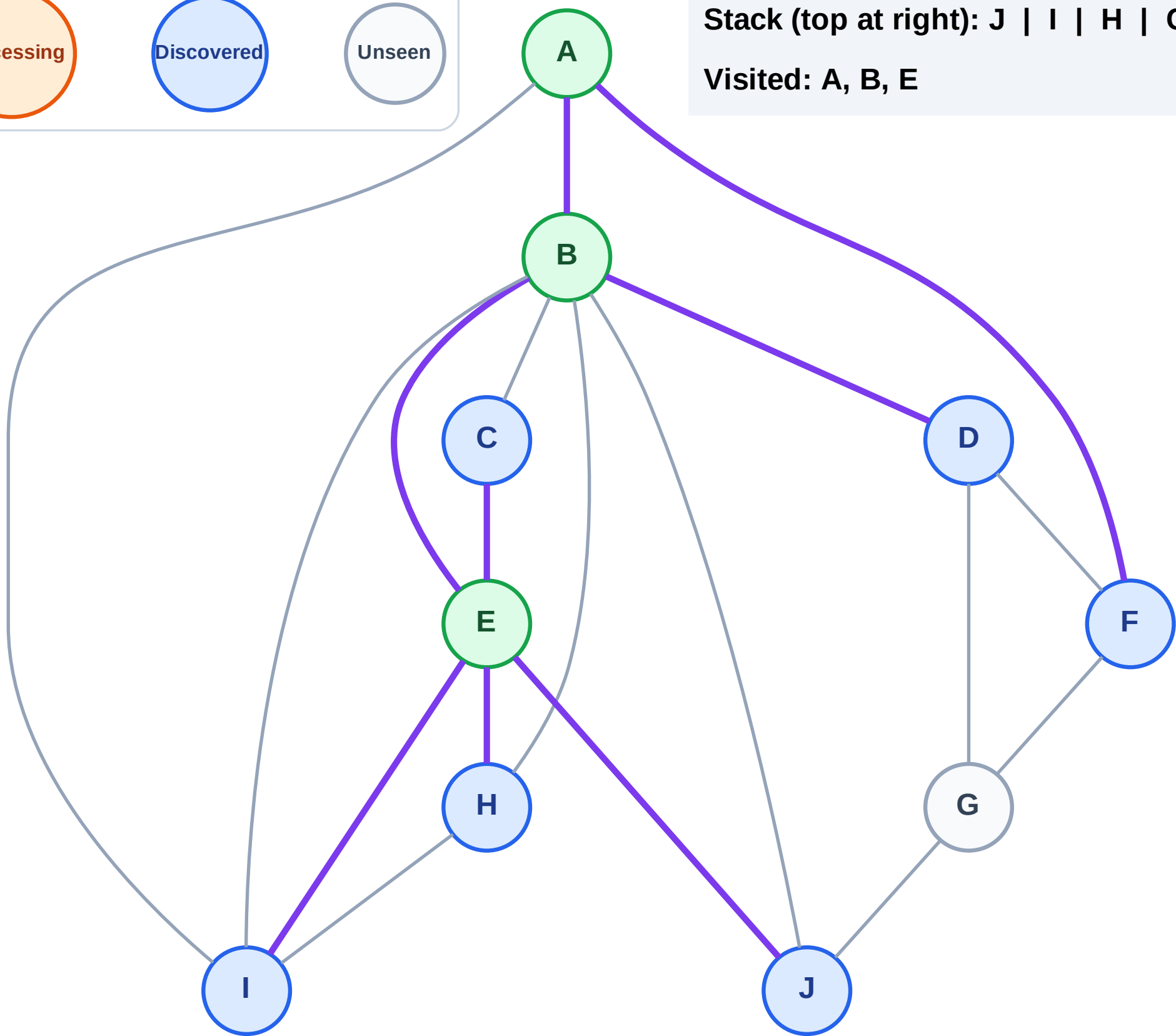


Step 7: Discover F from A

1. *Journal of the American Medical Association*, 2000; 283: 2689-2696.

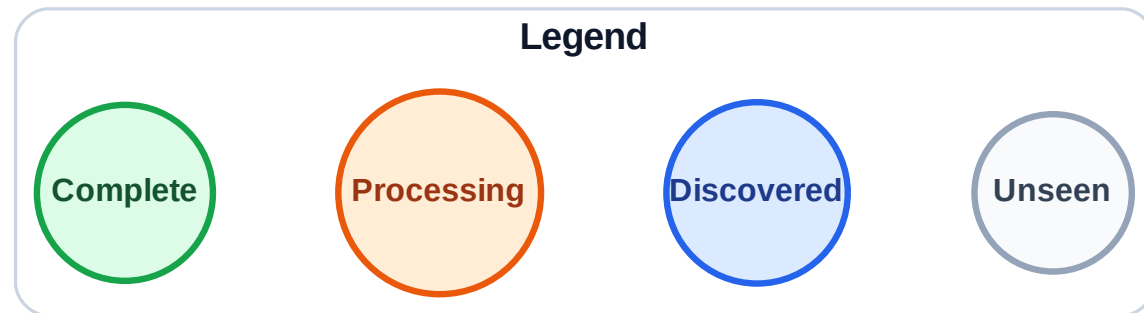
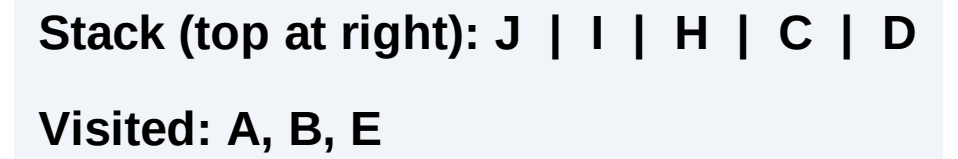


Visited: A, B, E



Step 8: Process node F

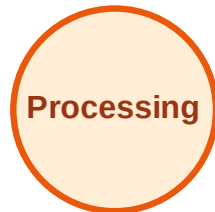
Step 8: Process node F



DFS Traversal

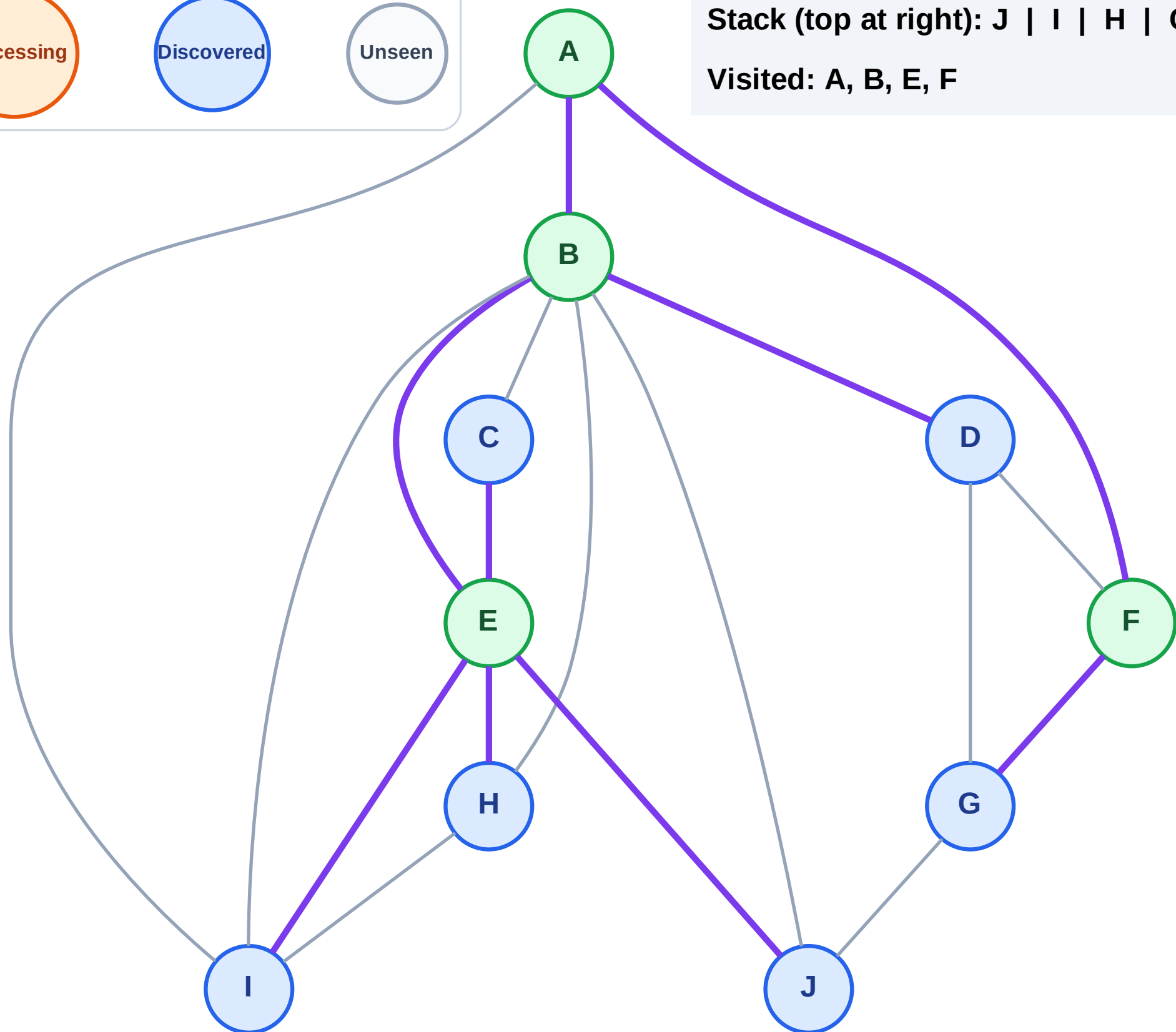
Step 9: Discover G from F

Legend



Stack (top at right): J | I | H | C | D | G

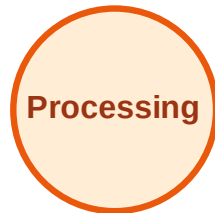
Visited: A, B, E, F



DFS Traversal

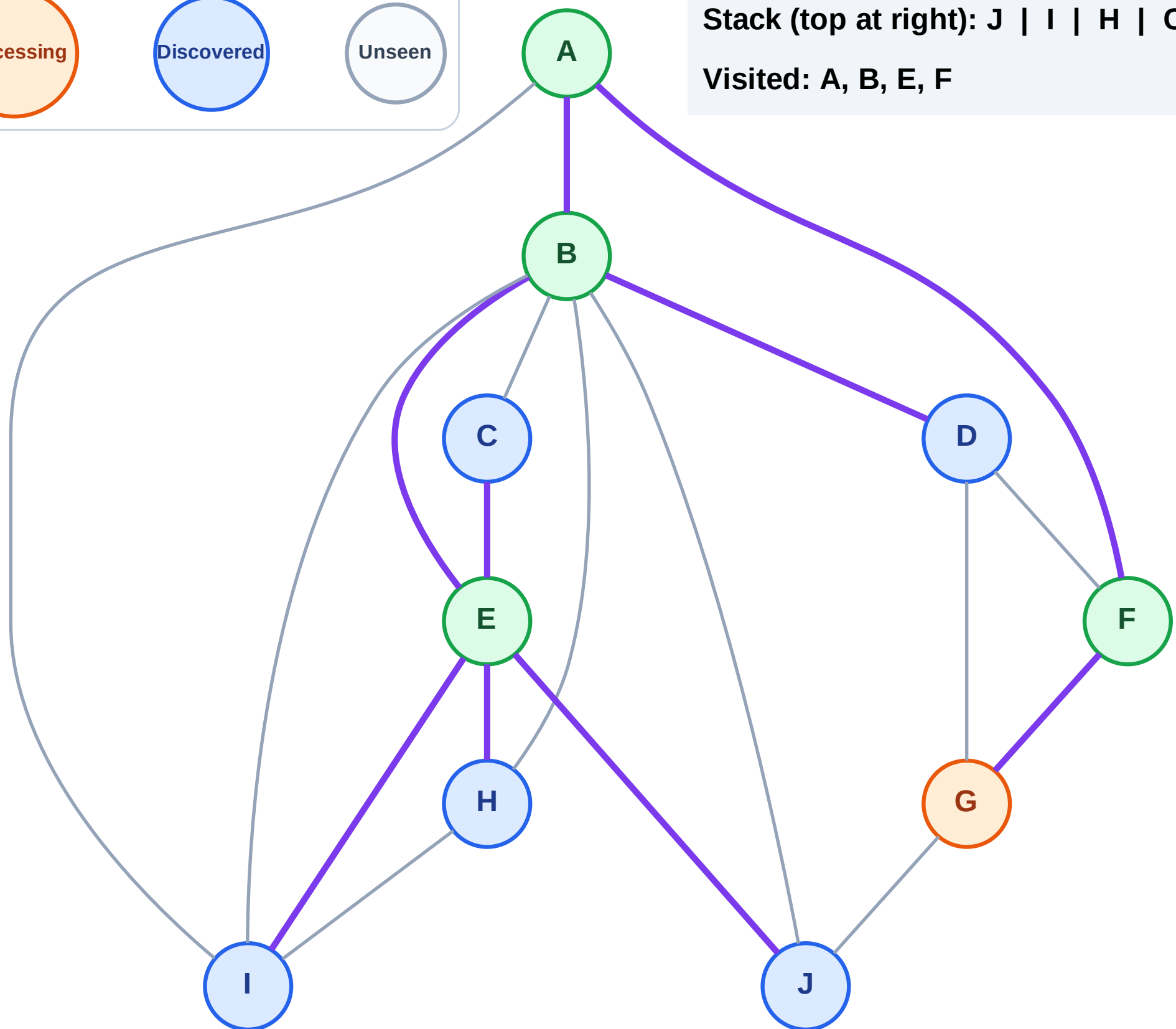
Step 10: Process node G

Legend



Stack (top at right): J | I | H | C | D

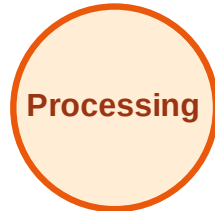
Visited: A, B, E, F



DFS Traversal

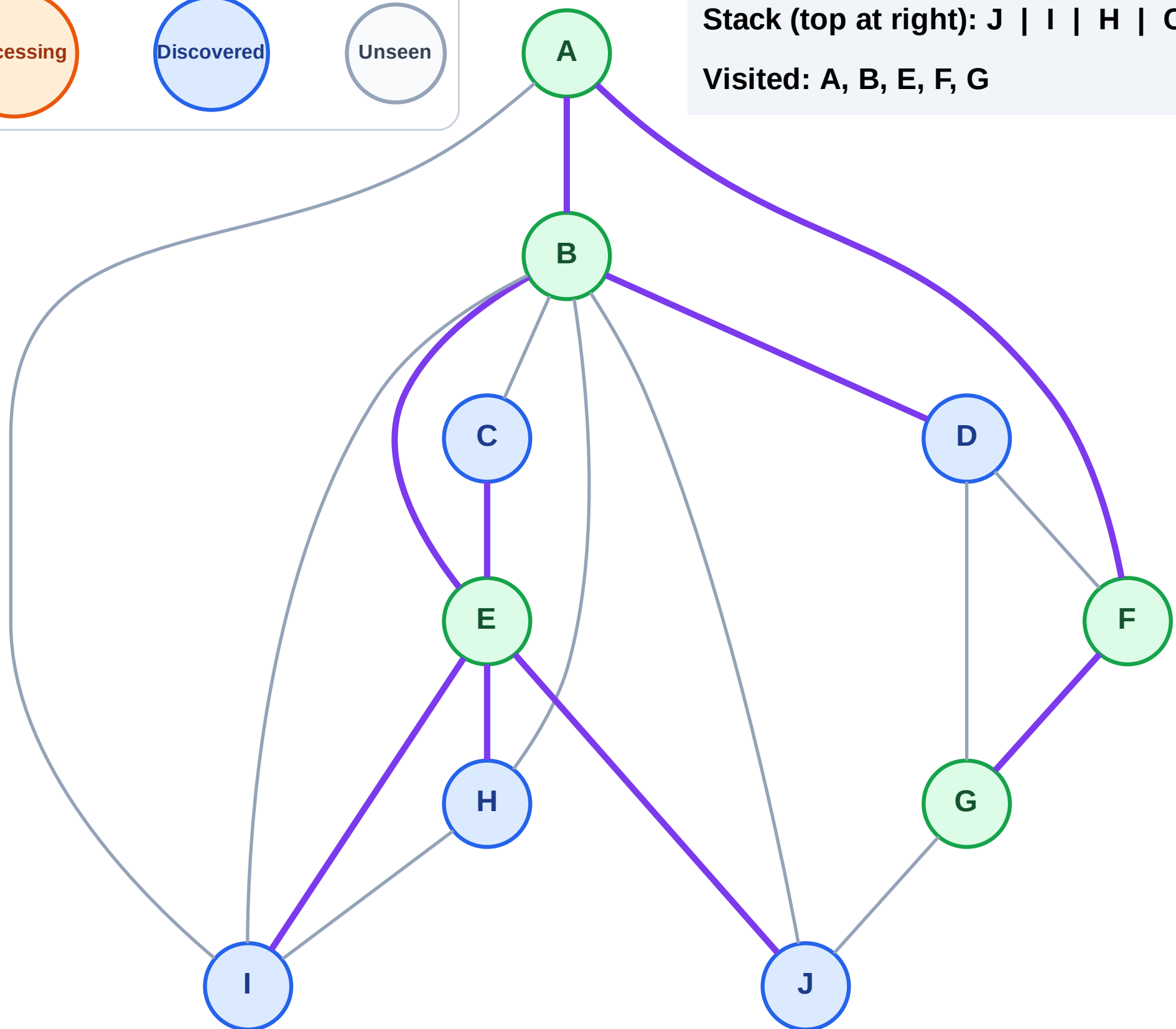
Step 11: No new neighbors from G

Legend



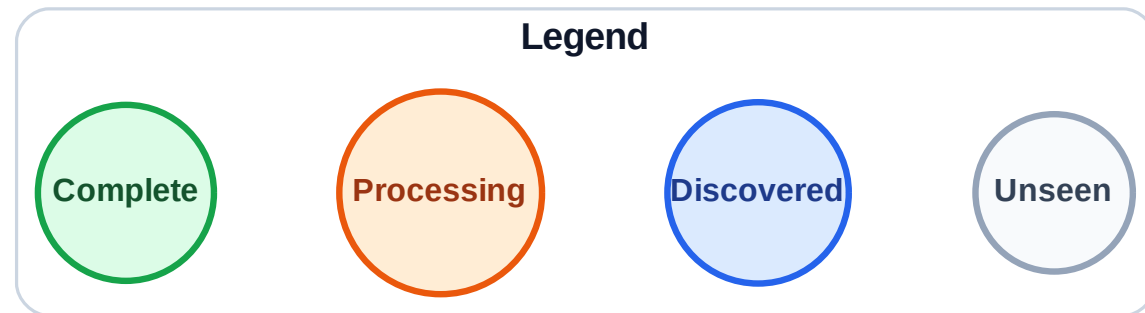
Stack (top at right): J | I | H | C | D

Visited: A, B, E, F, G



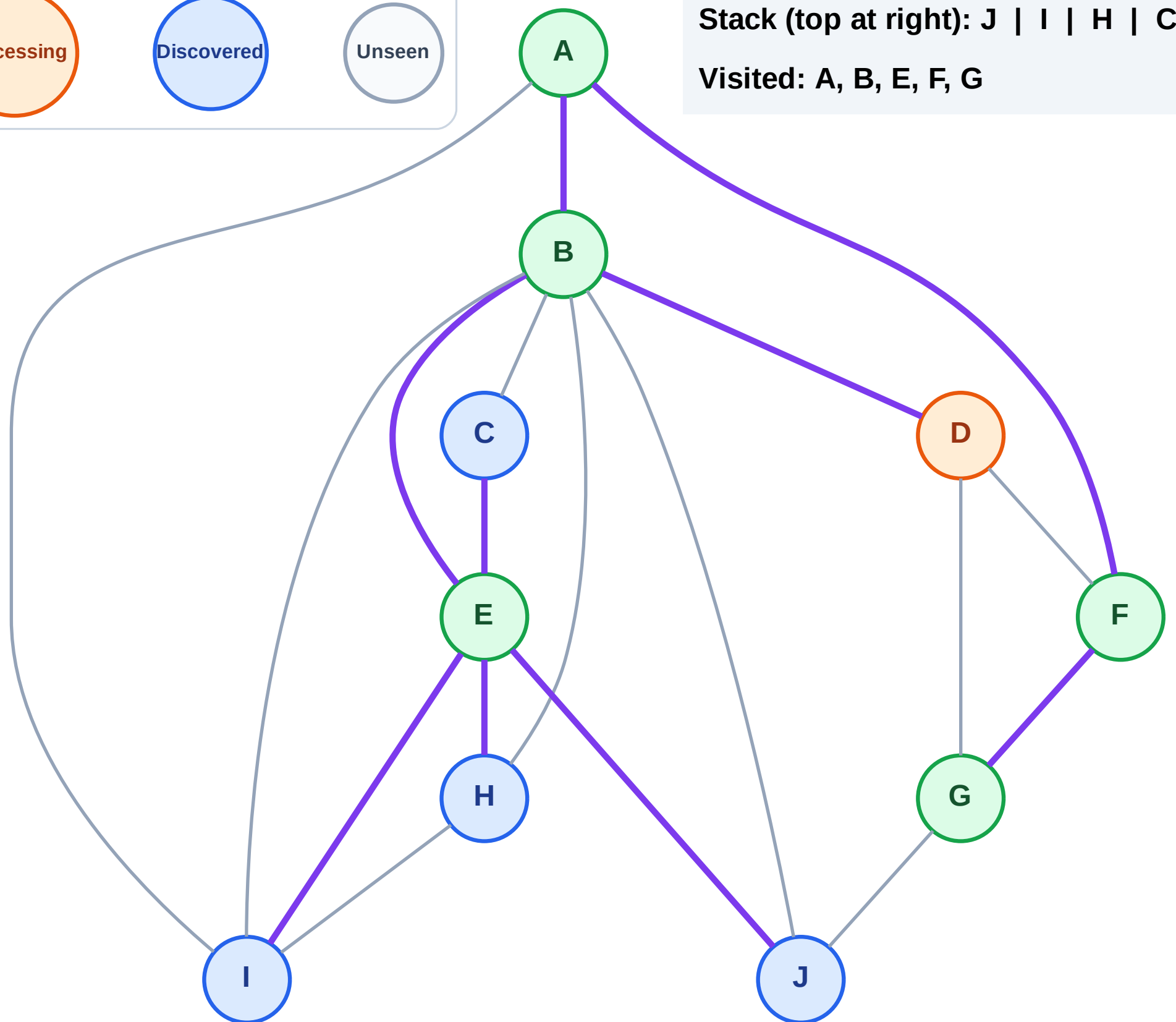
DFS Traversal

Step 12: Process node D



Stack (top at right): J | I | H | C

Visited: A, B, E, F, G



DFS Traversal

Step 13: No new neighbors from D

Legend

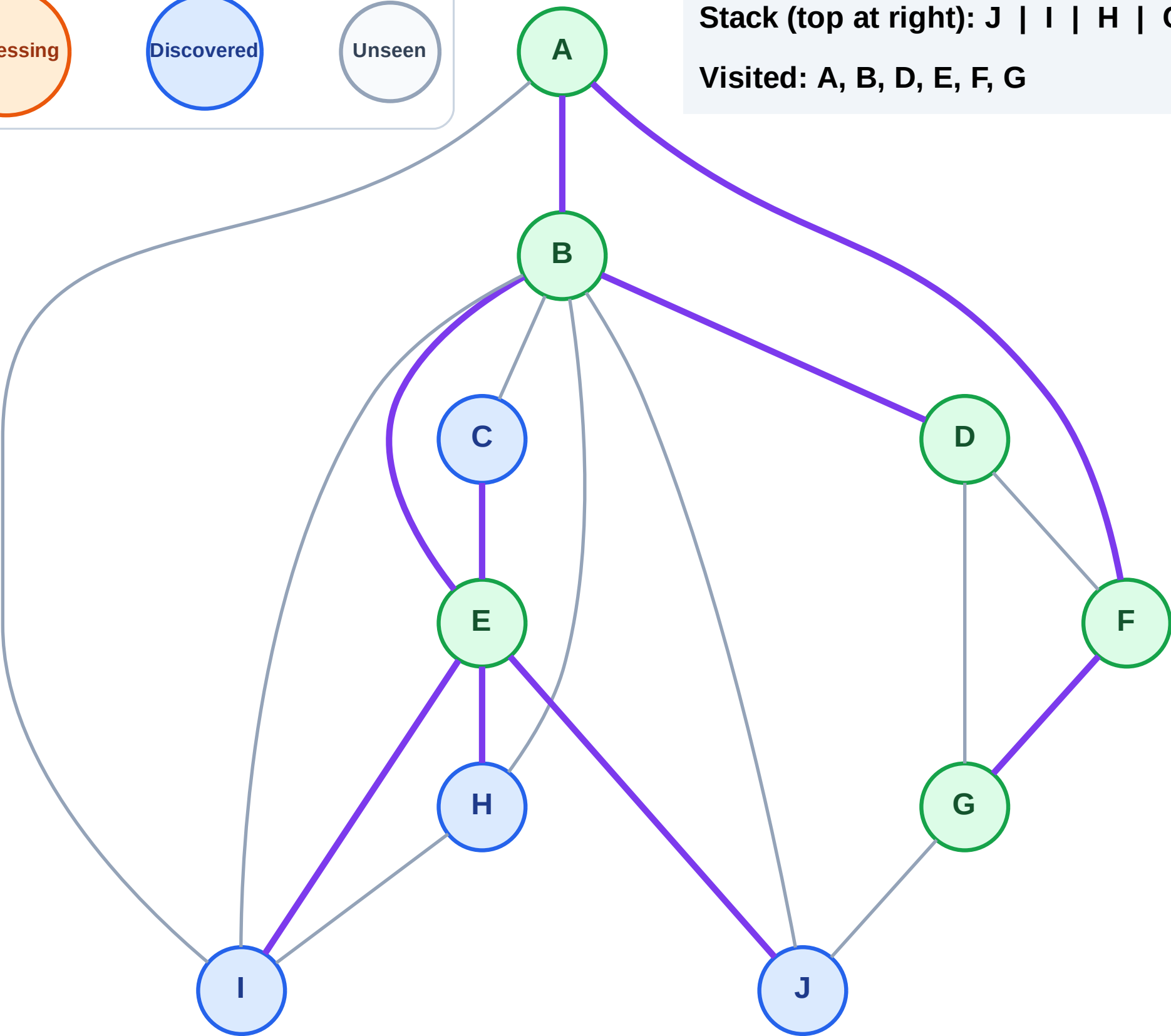
Complete

Processing

Discovered

Unseen

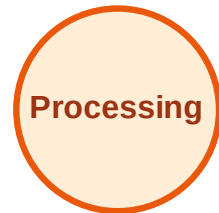
Stack (top at right): J | I | H | C
Visited: A, B, D, E, F, G



DFS Traversal

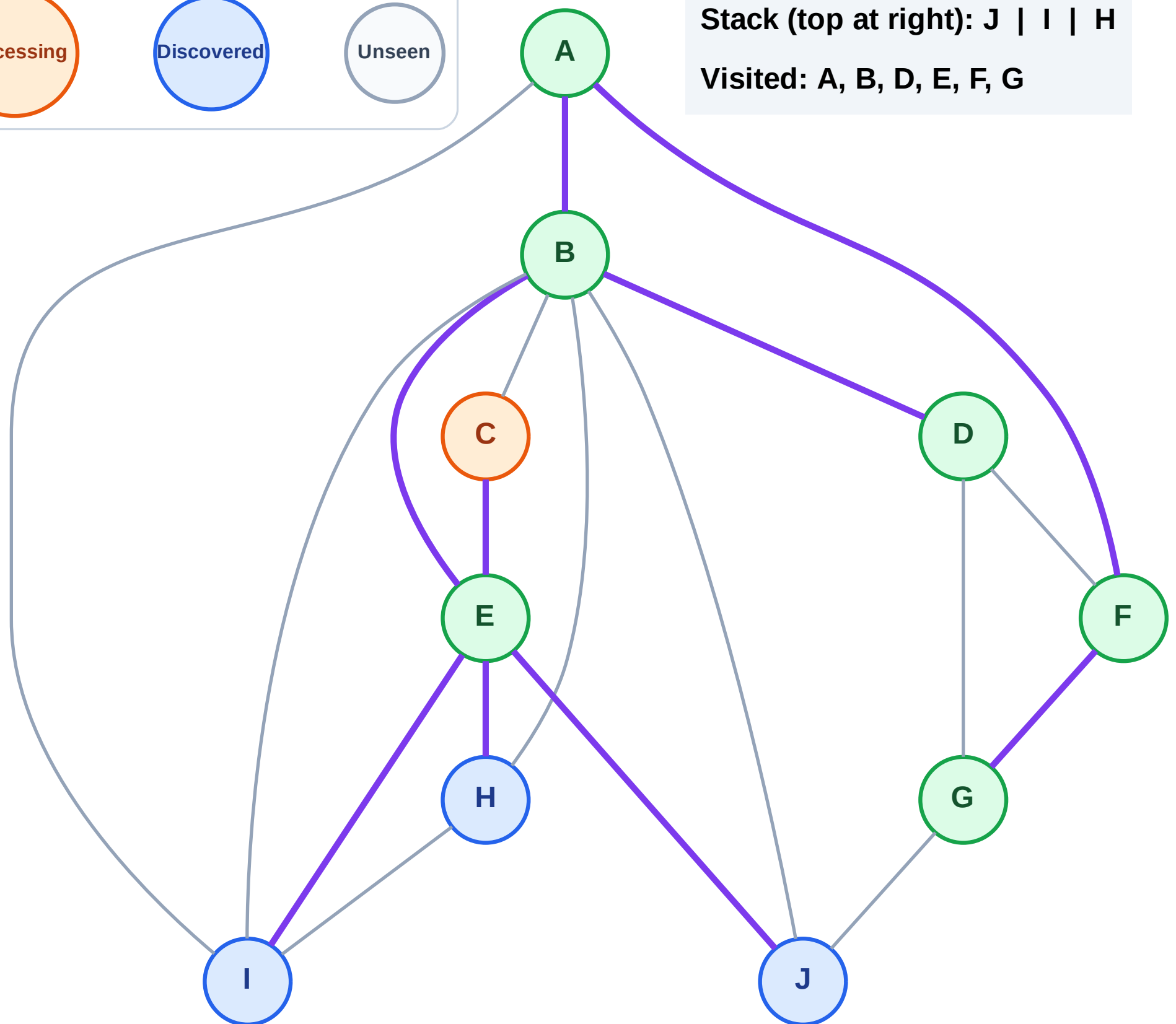
Step 14: Process node C

Legend



Stack (top at right): J | I | H

Visited: A, B, D, E, F, G



DFS Traversal

Step 15: No new neighbors from C

Legend

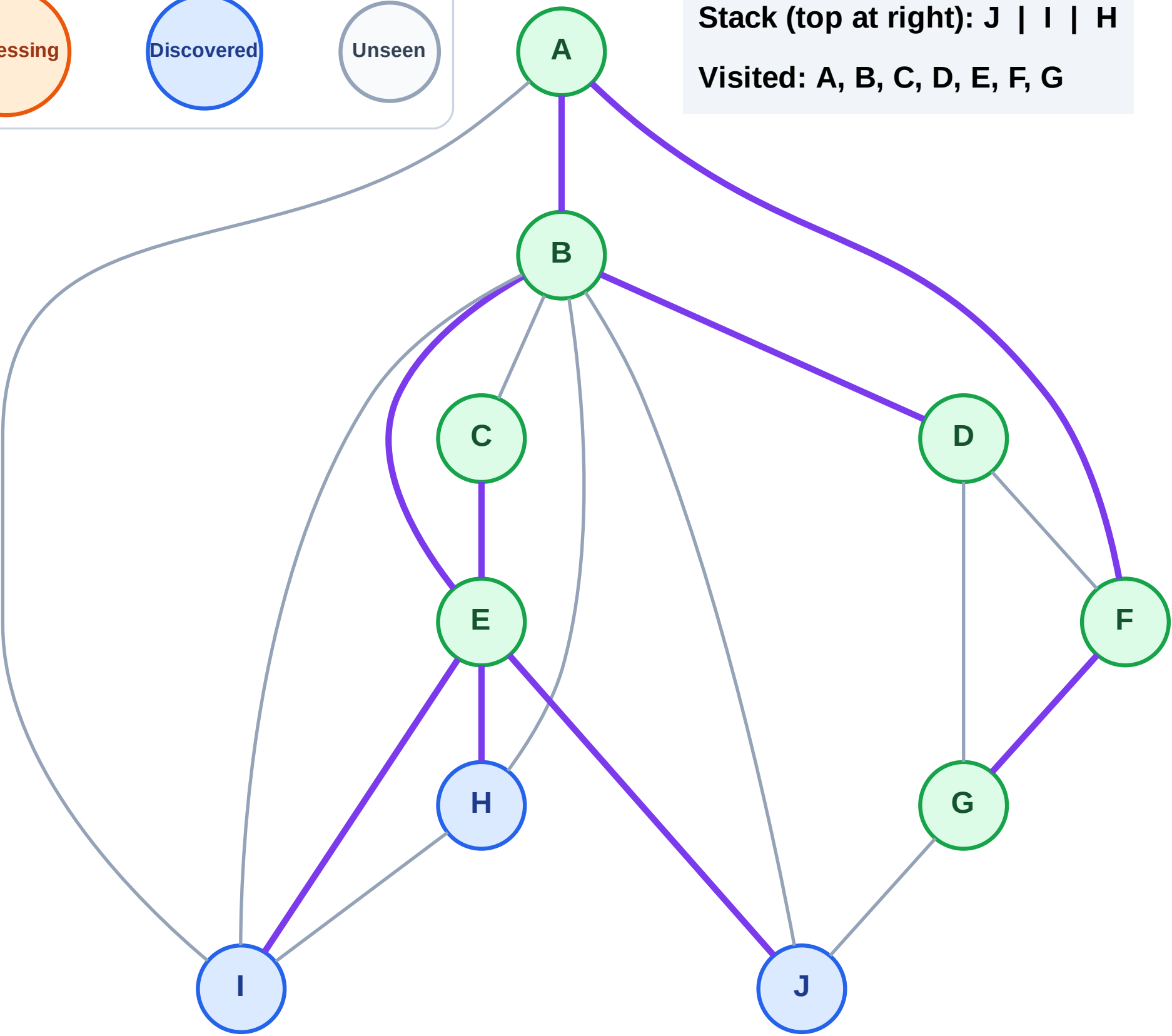
Complete

Processing

Discovered

Unseen

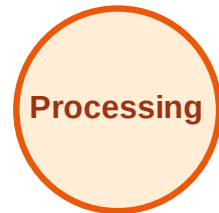
Stack (top at right): J | I | H
Visited: A, B, C, D, E, F, G



DFS Traversal

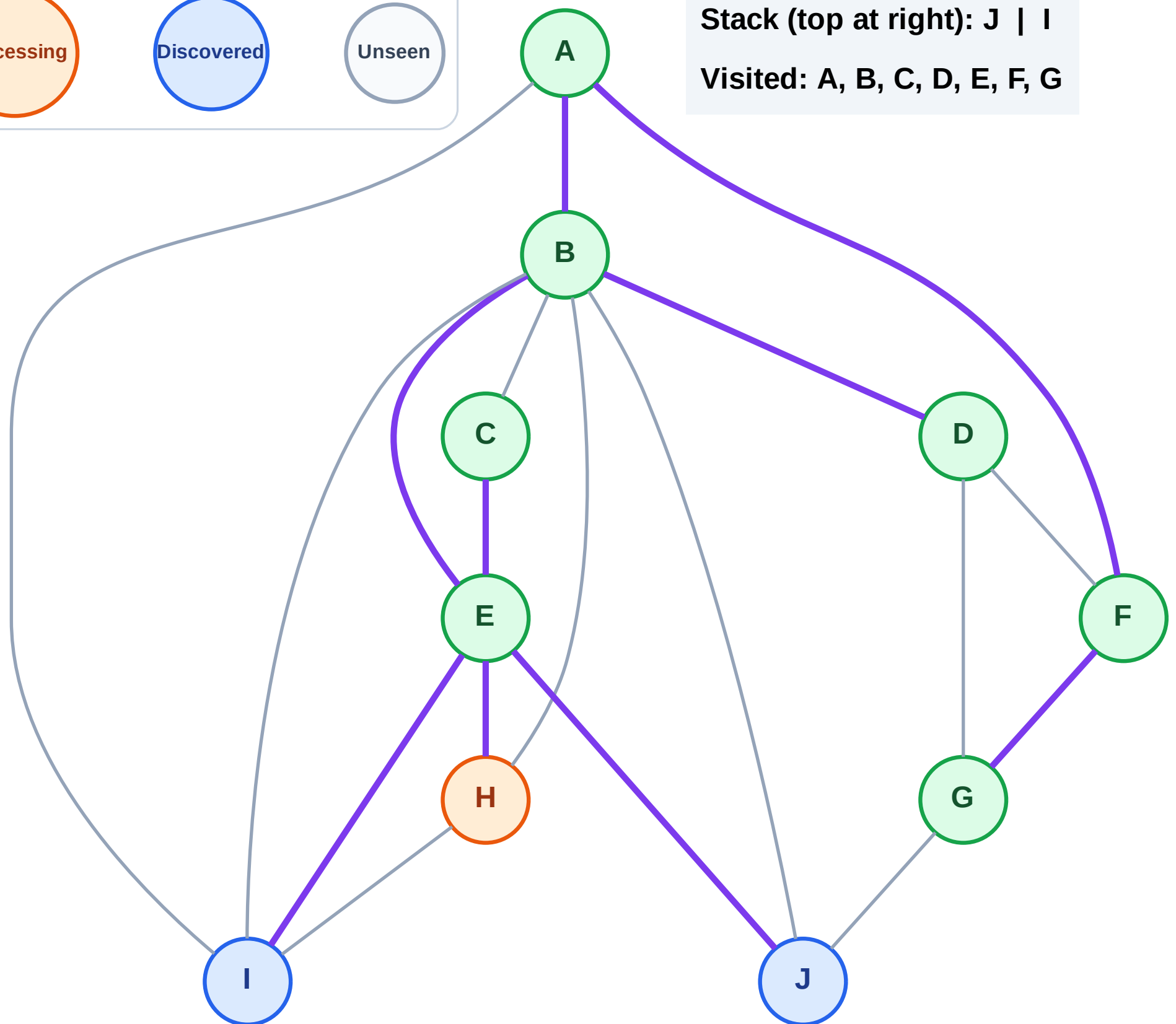
Step 16: Process node H

Legend



Stack (top at right): J | I

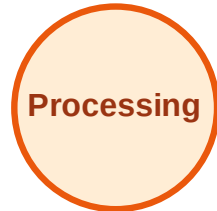
Visited: A, B, C, D, E, F, G



DFS Traversal

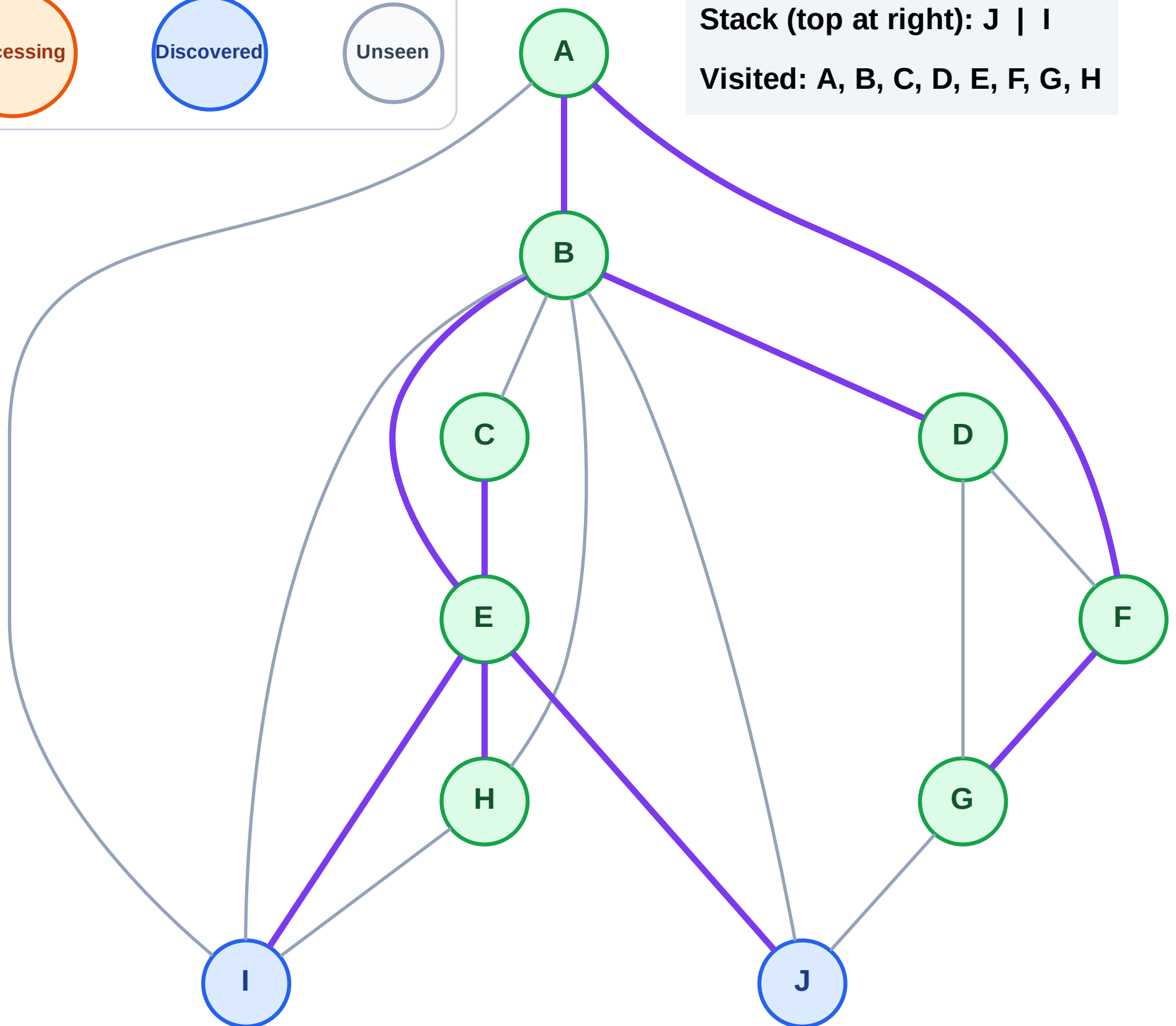
Step 17: No new neighbors from H

Legend



Stack (top at right): J | I

Visited: A, B, C, D, E, F, G, H



DFS Traversal

Step 18: Process node I

Legend

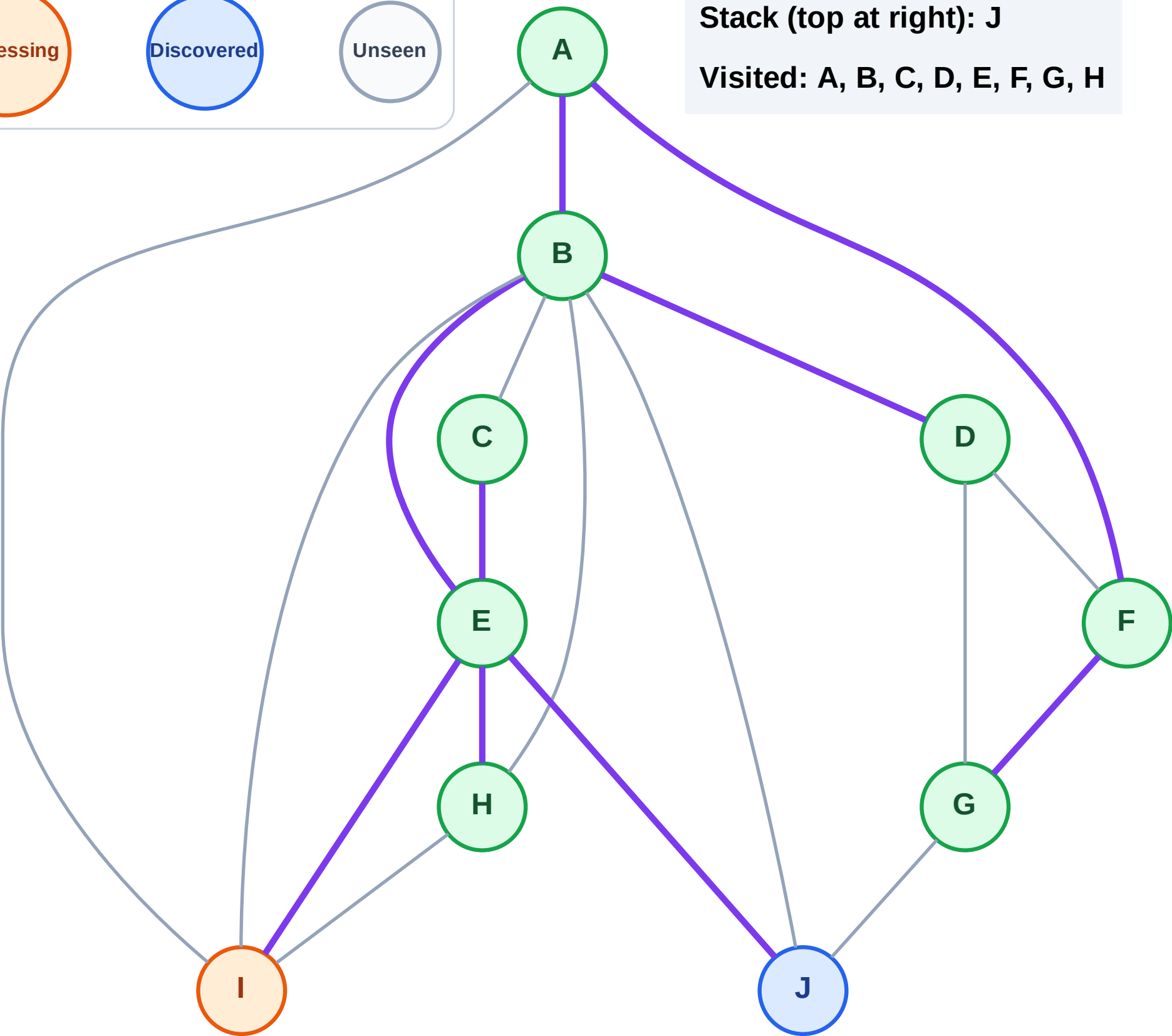
Complete

Processing

Discovered

Unseen

Stack (top at right): J
Visited: A, B, C, D, E, F, G, H



DFS Traversal

Step 19: No new neighbors from I

Legend

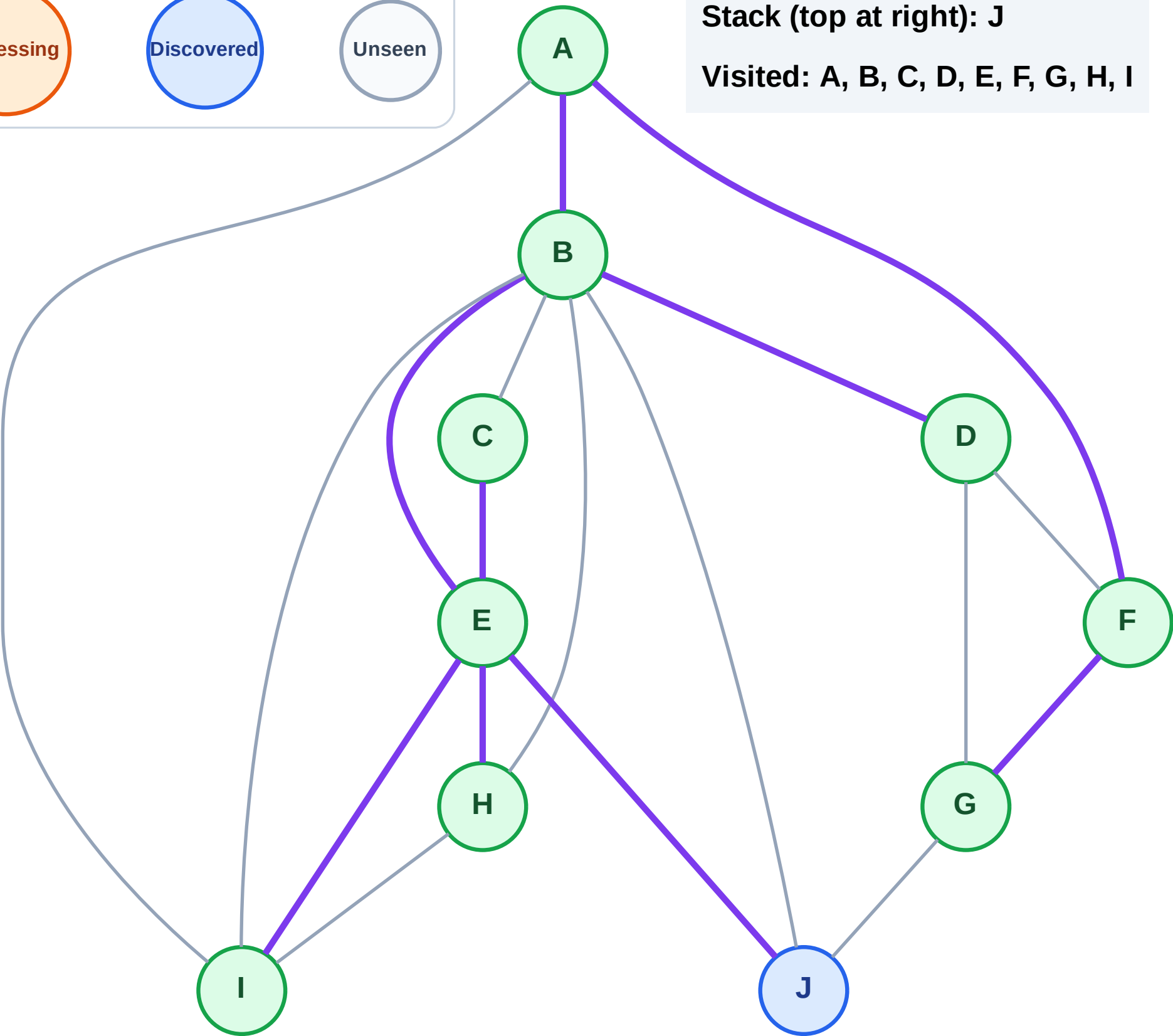
Complete

Processing

Discovered

Unseen

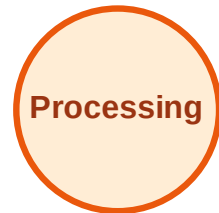
Stack (top at right): J
Visited: A, B, C, D, E, F, G, H, I



DFS Traversal

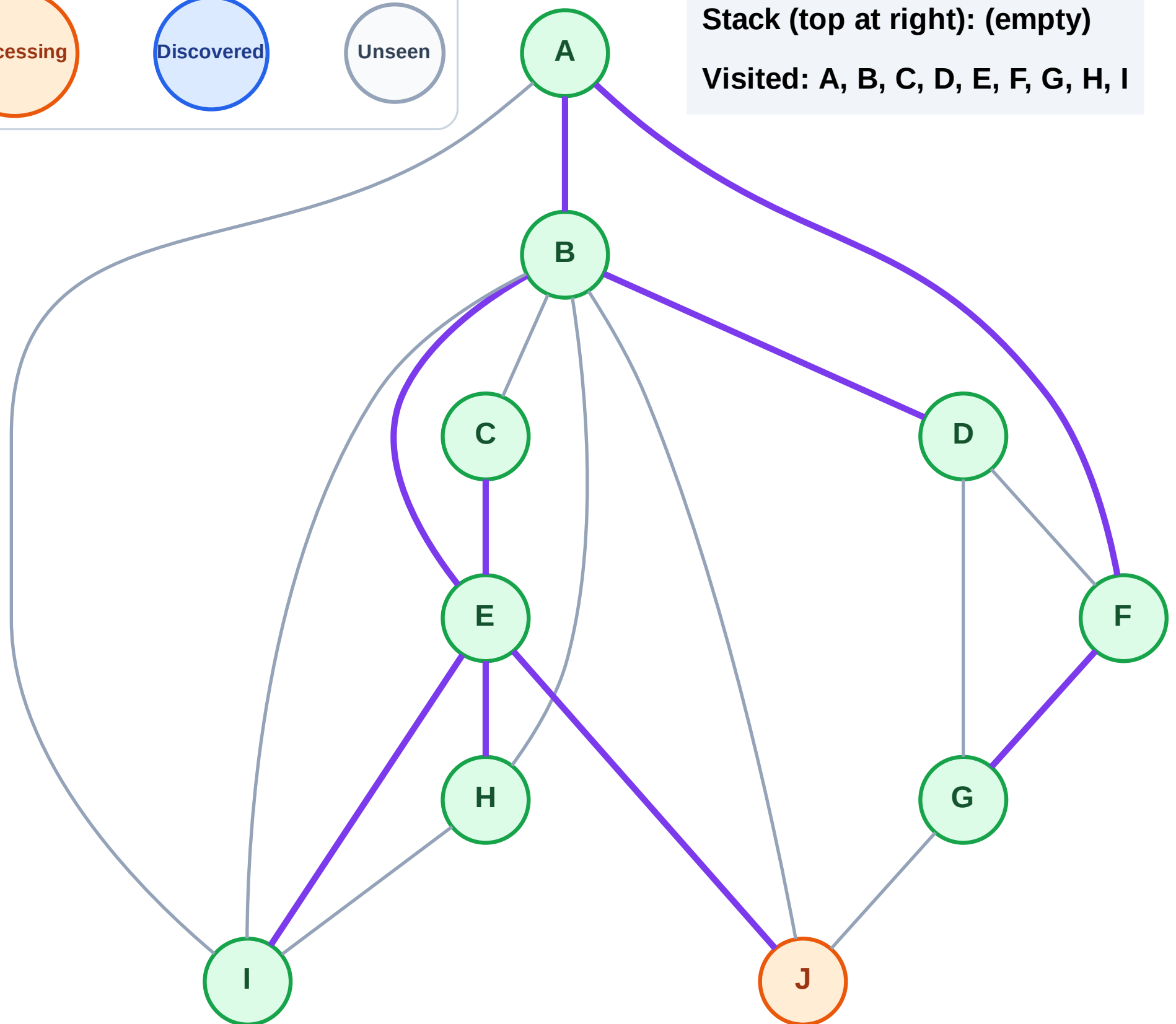
Step 20: Process node J

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I



DFS Traversal

Step 21: No new neighbors from J

Legend

Complete

Processing

Discovered

Unseen

Stack (top at right): (empty)
Visited: A, B, C, D, E, F, G, H, I, J

